

Applications to Analysis

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Differential Equations and Stability

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Differential equations and differential-algebraic equations arise in numerous branches of science and engineering that include biology, chemistry, medicine, structural mechanics, and electrical engineering. This chapter is concerned with linear differential(-algebraic) equations with constant coefficients that can be analyzed completely via techniques from linear algebra. We discuss the existence and uniqueness of solutions of such equations as well as the stability theory.

55.1 Linear Differential Equations with Constant Coefficients: Basic Concepts

Definitions:

A **linear differential equation** in an unknown function $x : \mathbb{R} \rightarrow \mathbb{C}$, $t \mapsto x(t)$, has the form

$$\dot{x} = ax + f,$$

where $a \in \mathbb{C}$ and the inhomogeneity $f : \mathbb{R} \rightarrow \mathbb{C}$ is a given function. Here $\dot{x}$ denotes the derivative of $x(t)$ with respect to t .

A **linear differential equation of order k** for an unknown function $x : \mathbb{R} \rightarrow \mathbb{C}$ has the form

$$a_k x^{(k)} + \cdots + a_0 x = f,$$

where $a_0, \dots, a_k \in \mathbb{C}$, $a_k \neq 0$ and $f : \mathbb{R} \rightarrow \mathbb{C}$. Here $x^{(k)}$ denotes the k -th derivative of $x(t)$ with respect to t .

A **system of linear differential(-algebraic) equations with constant coefficients** has the form

$$E \dot{x} = Ax + f,$$

where $E, A \in \mathbb{C}^{m \times n}$ are **coefficient matrices**, $x : \mathbb{R} \rightarrow \mathbb{C}^n$ is a vector-valued function of unknowns, and the inhomogeneity $f : \mathbb{R} \rightarrow \mathbb{C}^m$ is a given vector-valued function.

If $E = I_n$ and $A \in \mathbb{C}^{n \times n}$, then $E \dot{x} = Ax + f$ is a system of **ordinary differential equations**; otherwise it is a system of **differential-algebraic equations**.

A **homogeneous** system has $\mathbf{f}(t) \equiv 0$; otherwise the system is **inhomogeneous**.

A **system of linear differential-algebraic equations of order k** for an unknown function $\mathbf{x} : \mathbb{R} \rightarrow \mathbb{C}^n$ has the form $A_k \mathbf{x}^{(k)} + \cdots + A_0 \mathbf{x} = \mathbf{f}$, where $A_0, \dots, A_k \in \mathbb{C}^{m \times n}$, $A_k \neq 0$ and $\mathbf{f} : \mathbb{R} \rightarrow \mathbb{C}^m$.

A continuously differentiable function $\mathbf{x} : \mathbb{R} \rightarrow \mathbb{C}^n$ is a **solution** of $E \dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$ with a sufficiently often differentiable function $\mathbf{f}$ if it satisfies the equation pointwise.

A solution of $E \dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$ that also satisfies an initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$ with $t_0 \in \mathbb{R}$ and $\mathbf{x}_0 \in \mathbb{C}^n$ is a solution of the **initial value problem**.

If the initial value problem $E \dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$, $\mathbf{x}(t_0) = \mathbf{x}_0$ has a solution, then the initial condition is called **consistent**.

Facts:

1. [Cam80, p. 33] Let $E, A \in \mathbb{C}^{n \times n}$. If E is nonsingular, then the system $E \dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$ is equivalent to the system of ordinary differential equations $\dot{\mathbf{x}} = E^{-1}A\mathbf{x} + E^{-1}\mathbf{f}$.
2. [Arn92, p. 105] The k -th order system of differential-algebraic equations

$$A_k \mathbf{y}^{(k)} + \cdots + A_0 \mathbf{y} = \mathbf{g}$$

can be rewritten as a first-order system $E \dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$, where $\mathbf{f} = [0, \dots, 0, \mathbf{g}^T]^T$ and

$$E = \begin{bmatrix} I & & & \\ & \ddots & & \\ & & I & \\ & & & A_k \end{bmatrix}, \quad A = \begin{bmatrix} 0 & I & & 0 \\ \vdots & \ddots & \ddots & \\ 0 & \cdots & 0 & I \\ -A_0 & \cdots & -A_{k-2} & -A_{k-1} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} \mathbf{y} \\ \vdots \\ \mathbf{y}^{(k-2)} \\ \mathbf{y}^{(k-1)} \end{bmatrix}.$$

Applications:

1. Consider a mass–spring–damper model as shown in Figure 55.1. This model is described by the equation

$$m\ddot{x} + d\dot{x} + kx = 0,$$

where m is a mass, k is a spring constant, and d is a damping parameter. Since $m \neq 0$, we obtain the following first-order system of ordinary differential equations

$$\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k/m & -d/m \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix},$$

where the velocity is denoted by v .

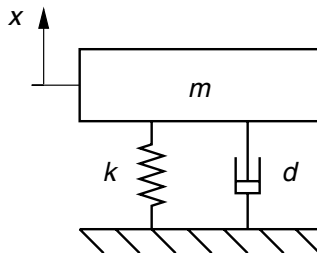


FIGURE 55.1 A mass–spring–damper model.

2. Consider a one-dimensional heat equation

$$\frac{\partial}{\partial t} T(t, \xi) = c \frac{\partial^2}{\partial \xi^2} T(t, \xi), \quad (t, \xi) \in (0, t_e) \times (0, l),$$

together with an initial condition $T(0, \xi) = g(\xi)$ and Cauchy boundary conditions

$$\alpha_1 T(t, 0) + \alpha_2 \frac{\partial}{\partial n} T(t, 0) = u(t),$$

$$\beta_1 T(t, l) + \beta_2 \frac{\partial}{\partial n} T(t, l) = v(t).$$

Here $T(t, \xi)$ is the temperature field in a thin beam of length l , $c > 0$ is the heat conductivity of the material, $g(\xi)$, $u(t)$, and $v(t)$ are given functions, and $\frac{\partial}{\partial n}$ denotes the derivative in the direction of the outward normal. A spatial discretization by a finite difference method with $n + 1$ equidistant grid points leads to the initial value problem $\dot{\mathbf{x}} = A_{a,b} \mathbf{x} + \mathbf{f}$, $\mathbf{x}(0) = \mathbf{x}_0$, where

$$\mathbf{x}(t) = [T(t, h), T(t, 2h), \dots, T(t, nh)]^T, \quad \mathbf{x}_0 = [g(h), g(2h), \dots, g(nh)]^T,$$

$$\mathbf{f}(t) = [cu(t)/(h^2\alpha_1 - h\alpha_2), 0, \dots, 0, cv(t)/(h^2\beta_1 + h\beta_2)]^T,$$

and

$$A_{a,b} = \frac{c}{h^2} \begin{bmatrix} -a & 1 & & & \\ & 1 & -2 & 1 & \\ & & \ddots & \ddots & \ddots \\ & & & 1 & -2 & 1 \\ & & & & 1 & -b \end{bmatrix} \in \mathbb{R}^{n \times n}$$

with $h = l/(n + 1)$, $a = (2h\alpha_1 - \alpha_2)/(h\alpha_1 - \alpha_2)$, and $b = (2h\beta_1 + \beta_2)/(h\beta_1 + \beta_2)$.

3. A simple pendulum as shown in Figure 55.2 describes the movement of a mass point with mass m and Cartesian coordinates (x, y) under the influence of gravity in a distance l around the origin.

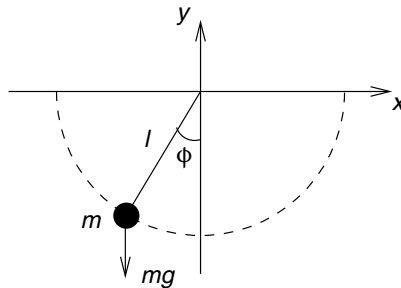


FIGURE 55.2 A simple pendulum.

The equations of motion have the form

$$\begin{aligned} m\ddot{x} + 2x\lambda &= 0, \\ m\ddot{y} + 2y\lambda + mg &= 0, \\ x^2 + y^2 - l^2 &= 0, \end{aligned}$$

where λ is a Lagrange multiplier. Transformation of this system into the first-order form by introducing new variables $v = \dot{x}$ and $w = \dot{y}$ and linearization at the equilibrium $x_e = 0$, $y_e = -l$, $v_e = 0$, $w_e = 0$, and $\lambda_e = mg/(2l)$ yields the homogeneous first-order linear differential-algebraic system

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & m & 0 & 0 \\ 0 & 0 & 0 & m & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{v} \\ \tilde{w} \\ \tilde{\lambda} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -2\lambda_e & 0 & 0 & 0 & 0 \\ 0 & -2\lambda_e & 0 & 0 & 2l \\ 0 & -2l & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{v} \\ \tilde{w} \\ \tilde{\lambda} \end{bmatrix},$$

where $\tilde{x} = x - x_e$, $\tilde{y} = y - y_e$, $\tilde{v} = v - v_e$, $\tilde{w} = w - w_e$, and $\tilde{\lambda} = \lambda - \lambda_e$.

The motion of the pendulum can also be described by the ordinary differential equation

$$\ddot{\varphi} = -\omega^2 \sin(\varphi),$$

where φ is an angle between the vertical axis and the pendulum and $\omega = \sqrt{g/l}$ is an angular frequency of the motion. By introducing a new variable $\psi = \dot{\varphi}$ and linearization at the equilibrium $\varphi_e = 0$ and $\psi_e = 0$, we obtain the first-order homogeneous system

$$\begin{bmatrix} \tilde{\dot{\varphi}} \\ \tilde{\dot{\psi}} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega^2 & 0 \end{bmatrix} \begin{bmatrix} \tilde{\varphi} \\ \tilde{\psi} \end{bmatrix}.$$

4. Consider a simple RLC electrical circuit as shown in Figure 55.3. Using Kirchoff's and Ohm's laws, the circuit can be described by the system $E\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$ with

$$E = \begin{bmatrix} L & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1/C & 0 & 0 & 0 \\ -R & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} i \\ v_L \\ v_C \\ v_R \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -v \end{bmatrix}.$$

Here R , L , and C are the resistance, inductance, and capacitance, respectively; v_R , v_L , and v_C are the corresponding voltage drops, i is the current, and v is the voltage source. From the last two equations

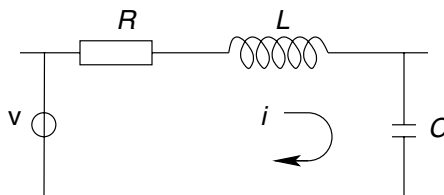


FIGURE 55.3 A simple RLC circuit.

in the system, we find $v_R = Ri$ and $v_L = v - v_C - Ri$. Substituting v_L in the first equation and introducing a new variable $w_C = i/C$, we obtain the system of ordinary differential equations

$$\begin{bmatrix} \dot{v}_C \\ \dot{w}_C \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1/(LC) & -R/L \end{bmatrix} \begin{bmatrix} v_C \\ w_C \end{bmatrix} + \begin{bmatrix} 0 \\ v/(LC) \end{bmatrix}.$$

This shows the relationship to the mass–spring–damper model as in Application 1.

55.2 Linear Ordinary Differential Equations

Facts:

The following facts can be found in [Gan59a, pp. 116–124, 153–154].

1. Let $J_A = T^{-1}AT$ be in Jordan canonical form. Then $e^{At} = Te^{J_A t}T^{-1}$. (See Chapter 6 and Chapter 11 for more information on the Jordan canonical form and the matrix exponential.)
2. Every solution of the homogeneous system $\dot{\mathbf{x}} = A\mathbf{x}$ has the form $\mathbf{x}(t) = e^{At}\mathbf{v}$ with $\mathbf{v} \in \mathbb{C}^n$.
3. The initial value problem $\dot{\mathbf{x}} = A\mathbf{x}$, $\mathbf{x}(t_0) = \mathbf{x}_0$ has the unique solution $\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{x}_0$.
4. The initial value problem $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$, $\mathbf{x}(t_0) = \mathbf{x}_0$ has a unique solution for every initial vector $\mathbf{x}_0$ and every continuous inhomogeneity $\mathbf{f}$. This solution is given by

$$\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{x}_0 + \int_{t_0}^t e^{A(t-\tau)}\mathbf{f}(\tau) d\tau.$$

Examples:

1. Let

$$A = \begin{bmatrix} 3 & 3 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & 1 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{f}(t) = \begin{bmatrix} -3t^2 - 3t \\ 2t \\ t^2 + t - 1 \end{bmatrix}.$$

For

$$T = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} \quad \text{and} \quad T^{-1} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

we have

$$J_A = T^{-1}AT = \left[\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 2 & 0 \\ \hline 0 & 0 & 0 \end{array} \right].$$

Then

$$\begin{aligned} e^{At} &= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} \left[\begin{array}{cc|c} e^{2t} & te^{2t} & 0 \\ 0 & e^{2t} & 0 \\ \hline 0 & 0 & 1 \end{array} \right] \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} (1+t)e^{2t} & (1+t)e^{2t} - 1 & te^{2t} \\ 0 & 1 & 0 \\ -te^{2t} & -te^{2t} & (1-t)e^{2t} \end{bmatrix}. \end{aligned}$$

Every solution of the homogeneous system $\dot{\mathbf{x}} = A\mathbf{x}$ has the form

$$\mathbf{x}(t) = e^{At}\mathbf{v} = \begin{bmatrix} ((1+t)v_1 + (1+t)v_2 + tv_3)e^{2t} - v_2 \\ v_2 \\ (-tv_1 - tv_2 + (1-t)v_3)e^{2t} \end{bmatrix}$$

with $\mathbf{v} = [v_1, v_2, v_3]^T$. The solution of the initial value problem $\dot{\mathbf{x}} = A\mathbf{x}$, $\mathbf{x}(0) = \mathbf{x}_0$ has the form

$$\mathbf{x}(t) = e^{At}\mathbf{x}_0 = \begin{bmatrix} (3+6t)e^{2t} - 2 \\ 2 \\ (3-6t)e^{2t} \end{bmatrix}.$$

The initial value problem $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$, $\mathbf{x}(0) = \mathbf{x}_0$ has the solution

$$\mathbf{x}(t) = \begin{bmatrix} (3+6t)e^{2t} + t - 2 \\ t^2 + 2 \\ (3-6t)e^{2t} + 1 \end{bmatrix}.$$

Applications:

1. Consider the matrix A from the mass–spring–damper example

$$A = \begin{bmatrix} 0 & 1 \\ -k/m & -d/m \end{bmatrix}.$$

The Jordan canonical form of A is given by $J_A = T^{-1}AT = \text{diag}(\lambda_1, \lambda_2)$, where

$$T = \begin{bmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{bmatrix}, \quad T^{-1} = \frac{1}{\lambda_2 - \lambda_1} \begin{bmatrix} \lambda_2 & -1 \\ -\lambda_1 & 1 \end{bmatrix}$$

and

$$\lambda_1 = \frac{-d - \sqrt{d^2 - 4km}}{2m}, \quad \lambda_2 = \frac{-d + \sqrt{d^2 - 4km}}{2m}$$

are the eigenvalues of A . We have

$$e^{At} = T \text{diag}(e^{\lambda_1 t}, e^{\lambda_2 t}) T^{-1} = \frac{1}{\lambda_2 - \lambda_1} \begin{bmatrix} \lambda_2 e^{\lambda_1 t} - \lambda_1 e^{\lambda_2 t} & e^{\lambda_2 t} - e^{\lambda_1 t} \\ \lambda_1 \lambda_2 (e^{\lambda_1 t} - e^{\lambda_2 t}) & \lambda_2 e^{\lambda_2 t} - \lambda_1 e^{\lambda_1 t} \end{bmatrix}.$$

The solution of the mass–spring–damper model with the initial conditions $x(0) = x_0$ and $v(0) = 0$ is given by

$$x(t) = \frac{x_0}{\lambda_2 - \lambda_1} (\lambda_2 e^{\lambda_1 t} - \lambda_1 e^{\lambda_2 t}), \quad v(t) = \frac{\lambda_1 \lambda_2 x_0}{\lambda_2 - \lambda_1} (e^{\lambda_1 t} - e^{\lambda_2 t}).$$

2. Since the matrix $A_{a,b}$ in the semidiscretized heat equation is symmetric, there exists an orthogonal matrix U such that $U^T A_{a,b} U = \text{diag}(\lambda_1, \dots, \lambda_n)$, where $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ are the eigenvalues of $A_{a,b}$. (See Chapter 7.2 and Chapter 45.) In this case $e^{A_{a,b}t} = U \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}) U^T$.

55.3 Linear Differential-Algebraic Equations

Definitions:

A **Drazin inverse** A^D of a matrix $A \in \mathbb{C}^{n \times n}$ is defined as the unique solution of the system of matrix equations

$$A^D A A^D = A^D, \quad A A^D = A^D A, \quad A^{k+1} A^D = A^k,$$

where k is a smallest nonnegative integer such that $\text{rank}(A^{k+1}) = \text{rank}(A^k)$.

Let $E, A \in \mathbb{C}^{m \times n}$. A pencil of the form

$$\lambda E - A = \text{diag}(\mathcal{L}_{n_1}, \dots, \mathcal{L}_{n_p}, \mathcal{M}_{m_1}, \dots, \mathcal{M}_{m_q}, \mathcal{J}_k, \mathcal{N}_s)$$

is called pencil in **Kronecker canonical form** if the block entries have the following properties: every entry $\mathcal{L}_{n_j} = \lambda L_{n_j} - R_{n_j}$ is a bidiagonal block of size $n_j \times (n_j + 1)$, $n_j \in \mathbb{N}$, where

$$L_{n_j} = \begin{bmatrix} 1 & 0 & & \\ & \ddots & \ddots & \\ & & 1 & 0 \end{bmatrix}, \quad R_{n_j} = \begin{bmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & 0 & 1 \end{bmatrix};$$

every entry $\mathcal{M}_{m_j} = \lambda L_{m_j}^T - R_{m_j}^T$ is a bidiagonal block of size $(m_j + 1) \times m_j$, $m_j \in \mathbb{N}$; the entry $\mathcal{J}_k = \lambda I_k - A_k$ is a block of size $k \times k$, $k \in \mathbb{N}$, where A_k is in Jordan canonical form; the entry $\mathcal{N}_s = \lambda N_s - I_s$ is a block of size $s \times s$, $s \in \mathbb{N}$, where $N_s = \text{diag}(N_{s_1}, \dots, N_{s_r})$; and

$$N_{s_j} = \begin{bmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{bmatrix}$$

is a nilpotent Jordan block with index of nilpotency s_j .

The numbers $n_1, \dots, n_p$ are called the **right Kronecker indices** of the pencil $\lambda E - A$.

The numbers $m_1, \dots, m_q$ are called the **left Kronecker indices** of the pencil $\lambda E - A$.

The number $\nu = \max_{1 \leq j \leq r} s_j$ is called the **index** of the pencil $\lambda E - A$.

A matrix pencil $\lambda E - A$ with $E, A \in \mathbb{C}^{m \times n}$ is called **regular**, if $m = n$ and $\det(\lambda E - A) \neq 0$ for some $\lambda \in \mathbb{C}$. Otherwise, the pencil is called **singular**.

Let $E, A \in \mathbb{C}^{m,n}$. Subspaces $W_l \subset \mathbb{C}^m$ and $W_r \subset \mathbb{C}^n$ are called **left** and **right reducing subspaces** of the pencil $\lambda E - A$ if $W_l = E W_r + A W_r$ and $\dim(W_l) = \dim(W_r) - p$, where p is the number of $\mathcal{L}_{n_j}$ blocks in the Kronecker canonical form.

Let $\lambda E - A$ be a regular pencil. Subspaces $W_l, W_r \subset \mathbb{C}^n$ are called **left** and **right deflating subspaces** of $\lambda E - A$ if $W_l = E W_r + A W_r$ and $\dim(W_l) = \dim(W_r)$.

Let $W_1, W_2 \subset \mathbb{C}^n$ be subspaces such that $W_1 \cap W_2 = \{0\}$ and $W_1 + W_2 = \mathbb{C}^n$. A matrix $P \in \mathbb{C}^{n,n}$ is called a **projection** onto W_1 along W_2 if $P^2 = P$, $\text{range}(P) = W_1$, and $\ker(P) = W_2$.

Let $\lambda E - A$ be a regular pencil. If $T_l, T_r \in \mathbb{C}^{n \times n}$ are nonsingular matrices such that $T_l^{-1}(\lambda E - A)T_r$ is in Kronecker canonical form, then

$$P_l = T_l \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} T_l^{-1}, \quad P_r = T_r \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} T_r^{-1}$$

are the **spectral projections** onto the left and right deflating subspaces of $\lambda E - A$ corresponding to the finite eigenvalues along the left and right deflating subspaces corresponding to the eigenvalue at infinity.

Facts:

1. [Cam80, p. 8] If $A \in \mathbb{C}^{n \times n}$ is nonsingular, then $A^D = A^{-1}$.
2. [Cam80, p. 8] Let

$$J_A = T^{-1}AT = \begin{bmatrix} A_1 & \mathbf{0} \\ \mathbf{0} & A_0 \end{bmatrix}$$

be in Jordan canonical form, where A_1 contains all the Jordan blocks associated with the nonzero eigenvalues, and A_0 contains all the Jordan blocks associated with the eigenvalue 0. Then

$$A^D = T \begin{bmatrix} A_1^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} T^{-1}.$$

3. [Gan59b, pp. 29–37] For every matrix pencil $\lambda E - A$ with $E, A \in \mathbb{C}^{m \times n}$ there exist nonsingular matrices $T_l \in \mathbb{C}^{m \times m}$ and $T_r \in \mathbb{C}^{n \times n}$ such that $T_l^{-1}(\lambda E - A)T_r$ is in Kronecker canonical form. The Kronecker canonical form is unique up to permutation of the diagonal blocks, i.e., the kind, size, and number of the blocks are characteristic for the pencil $\lambda E - A$. (For more information on matrix pencils, see Section 43.1.)
4. [Gan59b, p. 47] If $\mathbf{f}(t) = [f_1(t), \dots, f_{n_j}(t)]^T$ is an n_j -times continuously differentiable vector-valued function and $g(t)$ is an arbitrary $(n_j + 1)$ -times continuously differentiable function, then the system $L_{n_j}\dot{\mathbf{x}} = R_{n_j}\mathbf{x} + \mathbf{f}$ has a continuously differentiable solution of the form

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_{n_j+1}(t) \end{bmatrix} = \begin{bmatrix} g(t) \\ g^{(1)}(t) - f_1(t) \\ \vdots \\ g^{(n_j)}(t) - \sum_{i=1}^{n_j} f_i^{(n_j-i)}(t) \end{bmatrix}.$$

A consistent initial condition has to satisfy this defining equation at t_0 .

5. [Gan59b, p. 47] A system of differential-algebraic equations $L_{m_j}^T \dot{\mathbf{x}} = R_{m_j}^T \mathbf{x} + \mathbf{f}$ with a vector-valued function $\mathbf{f}(t) = [f_1(t), \dots, f_{m_j+1}(t)]^T$ has a unique solution if and only if $\mathbf{f}$ is m_j -times continuously differentiable and $\sum_{i=1}^{m_j+1} f_i^{(i-1)}(t) \equiv 0$. If this holds, then the solution is given by

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_{m_j}(t) \end{bmatrix} = \begin{bmatrix} -\sum_{i=1}^{m_j} f_{i+1}^{(i-1)}(t) \\ \vdots \\ -f_{m_j+1}(t) \end{bmatrix}.$$

A consistent initial condition has to satisfy this defining equation at t_0 .

6. [Gan59b, p. 48] A system $N_{s_j}\dot{\mathbf{x}} = \mathbf{x} + \mathbf{f}$ has a unique continuously differentiable solution $\mathbf{x}$ if $\mathbf{f}$ is s_j -times continuously differentiable. This solution is given by

$$\mathbf{x}(t) = -\sum_{i=0}^{s_j-1} N_{s_j}^i \mathbf{f}^{(i)}(t).$$

A consistent initial condition has to satisfy this defining equation at t_0 .

7. [Cam80, pp. 37–39] If the pencil $\lambda E - A$ is regular of index ν , then for every ν -times differentiable inhomogeneity $\mathbf{f}$ there exists a solution of the differential-algebraic system $E\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$. Every solution of this system has the form

$$\mathbf{x}(t) = e^{\hat{E}^D \hat{A}(t-t_0)} \hat{E}^D \hat{E} \mathbf{v} + \int_{t_0}^t e^{\hat{E}^D \hat{A}(t-\tau)} \hat{E}^D \hat{\mathbf{f}}(\tau) d\tau - (I - \hat{E}^D \hat{E}) \sum_{j=0}^{\nu-1} (\hat{E} \hat{A}^D)^j \hat{A}^D \hat{\mathbf{f}}^{(j)}(t),$$

where $\mathbf{v} \in \mathbb{C}^n$, $\hat{E} = (\lambda_0 E - A)^{-1} E$, $\hat{A} = (\lambda_0 E - A)^{-1} A$, and $\hat{\mathbf{f}} = (\lambda_0 E - A)^{-1} \mathbf{f}$ for some $\lambda_0 \in \mathbb{C}$ such that $\lambda_0 E - A$ is nonsingular.

8. [Cam80, pp. 37–39] If the pencil $\lambda E - A$ is regular of index ν and if $\mathbf{f}$ is ν -times differentiable, then the initial value problem $E\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$, $\mathbf{x}(t_0) = \mathbf{x}_0$ possesses a solution if and only if there exists $\mathbf{v} \in \mathbb{C}^n$ that satisfies

$$\mathbf{x}_0 = \hat{E}^D \hat{E} \mathbf{v} - (I - \hat{E}^D \hat{E}) \sum_{j=0}^{\nu-1} (\hat{E} \hat{A}^D)^j \hat{A}^D \hat{\mathbf{f}}^{(j)}(t_0).$$

If such a $\mathbf{v}$ exists, then the solution is unique.

9. [KM06, p. 21] The existence of a unique solution of $E\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$, $\mathbf{x}(t_0) = \mathbf{x}_0$ does not imply that the pencil $\lambda E - A$ is regular.
10. [Cam80, pp. 41–44] If the pencil $\lambda E - A$ is singular, then the initial value problem $E\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$, $\mathbf{x}(t_0) = \mathbf{x}_0$ may have no solutions or the solution, if it exists, may not be unique.
11. [Sty02, pp. 23–26] Let the pencil $\lambda E - A$ be regular of index ν . If

$$T_l^{-1}(\lambda E - A)T_r = \begin{bmatrix} \lambda I - A_k & \mathbf{0} \\ \mathbf{0} & \lambda N_s - I \end{bmatrix}$$

is in Kronecker canonical form, then the solution of the initial value problem $E\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$, $\mathbf{x}(t_0) = \mathbf{x}_0$ can be represented as

$$\mathbf{x}(t) = \mathcal{F}(t - t_0) E \mathbf{x}_0 + \int_{t_0}^t \mathcal{F}(t - \tau) \mathbf{f}(\tau) d\tau + \sum_{j=0}^{\nu-1} F_{-j-1} \mathbf{f}^{(j)}(t),$$

where

$$\mathcal{F}(t) = T_r \begin{bmatrix} e^{A_k t} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} T_l^{-1}, \quad F_{-j} = T_r \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -N_s^{j-1} \end{bmatrix} T_l^{-1}.$$

Examples:

1. The system

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \dot{\mathbf{x}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ g(t) \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

has no solution if $g(t) \not\equiv 0$. For $g(t) \equiv 0$, this system has the solution $\mathbf{x}(t) = [e^t, \phi(t)]^T$, where $\phi(t)$ is a differentiable function such that $\phi(0) = 0$.

2. The system

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \dot{\mathbf{x}} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} -\sin(t) \\ -\cos(t) \\ 0 \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

has a unique solution $\mathbf{x}(t) = [\cos(t), 0]^T$, but the pencil $\lambda E - A$ is singular.

Applications:

1. The pencil in the linearized pendulum example has the Kronecker canonical form

$$\text{diag}(\mathcal{J}_2, \mathcal{N}_3) = \lambda \left[\begin{array}{cc|ccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] - \left[\begin{array}{cc|ccc} -i\sqrt{g/l} & 0 & 0 & 0 & 0 \\ 0 & i\sqrt{g/l} & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right].$$

This pencil is regular of index 3. Since the linearized pendulum system is homogeneous, it has a unique solution for every consistent initial condition.

2. The pencil of the circuit equation has the Kronecker canonical form

$$\text{diag}(\mathcal{J}_2, \mathcal{N}_2) = \lambda \left[\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] - \left[\begin{array}{cc|cc} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right],$$

with

$$\lambda_1 = -\frac{R}{2L} - \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}, \quad \lambda_2 = -\frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}.$$

This pencil is regular of index 1. Hence, there exists a unique continuous solution for every continuous voltage source $v(t)$ and for every consistent initial condition.

55.4 Stability of Linear Ordinary Differential Equations

The notion of stability is used to study the behavior of dynamical systems under initial perturbations around equilibrium points. In this section, we consider the stability of linear homogeneous ordinary differential equations with constant coefficients only. For extensions of this concept to general nonlinear systems, see, e.g., [Ces63] and [Hah67].

Definitions:

The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $\dot{\mathbf{x}} = A\mathbf{x}$ is called **stable in the sense of Lyapunov**, or simply **stable**, if for every $\varepsilon > 0$ there exists a $\delta = \delta(\varepsilon) > 0$ such that any solution $\mathbf{x}$ of $\dot{\mathbf{x}} = A\mathbf{x}$, $\mathbf{x}(t_0) = \mathbf{x}_0$ with $\|\mathbf{x}_0\|_2 < \delta$ satisfies $\|\mathbf{x}(t)\|_2 < \varepsilon$ for all $t \geq t_0$.

The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $\dot{\mathbf{x}} = A\mathbf{x}$ is called **asymptotically stable** if it is stable and $\lim_{t \rightarrow \infty} \mathbf{x}(t) = 0$ for any solution $\mathbf{x}$ of $\dot{\mathbf{x}} = A\mathbf{x}$.

The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $\dot{\mathbf{x}} = A\mathbf{x}$ is called **unstable** if it is not stable.

The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $\dot{\mathbf{x}} = A\mathbf{x}$ is called **exponentially stable** if there exist $\alpha > 0$ and $\beta > 0$ such that the solution $\mathbf{x}$ of $\dot{\mathbf{x}} = A\mathbf{x}$, $\mathbf{x}(t_0) = \mathbf{x}_0$ satisfies $\|\mathbf{x}(t)\|_2 \leq \alpha e^{-\beta(t-t_0)} \|\mathbf{x}_0\|_2$ for all $t \geq t_0$.

Facts:

1. [Gan59a, pp. 125–129] The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $\dot{\mathbf{x}} = A\mathbf{x}$ is stable if and only if all the eigenvalues of A have nonpositive real part and those with zero real part have the same algebraic and geometric multiplicities. If at least one of these conditions is violated, then the equilibrium $\mathbf{x}_e(t) \equiv 0$ of $\dot{\mathbf{x}} = A\mathbf{x}$ is unstable.
2. [Gan59a, pp. 125–129] The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable if and only if all the eigenvalues of A have negative real part.

3. [Ces63, p. 22] Let $p_A(\lambda) = \det(\lambda I - A) = \lambda^n + a_1\lambda^{n-1} + \cdots + a_n$ be the characteristic polynomial of $A \in \mathbb{R}^{n,n}$. If the equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable, then $a_j > 0$ for $j = 1, \dots, n$.
4. [Gan59b, pp. 185–189] The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable if and only if the Lyapunov equation $A^*X + XA = -Q$ has a unique Hermitian, positive definite solution X for every Hermitian, positive definite matrix Q .
5. [God97] Let H be a Hermitian, positive definite solution of the Lyapunov equation

$$A^*H + HA = -I$$

and let $\mathbf{x}$ be a solution of the initial value problem $\dot{\mathbf{x}} = A\mathbf{x}$, $\mathbf{x}(0) = \mathbf{x}_0$. Then in terms of the original data,

$$\|\mathbf{x}(t)\|_2 \leq \sqrt{\kappa(A)} e^{-t\|A\|_2/\kappa(A)} \|\mathbf{x}_0\|_2,$$

where $\kappa(A) = 2\|A\|_2\|H\|_2$.

6. [Hah67, pp. 113–117] The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $\dot{\mathbf{x}} = A\mathbf{x}$ is exponentially stable if and only if it is asymptotically stable.
7. [Hah67, p. 16] If the equilibrium $\mathbf{x}_e(t) \equiv 0$ of the homogeneous system $\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable, then all the solutions of the inhomogeneous system $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$ with a bounded inhomogeneity $\mathbf{f}$ are bounded.

Examples:

1. Consider the linear system $\dot{\mathbf{x}} = A\mathbf{x}$ with

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}.$$

For the initial condition $\mathbf{x}(0) = [1, 0]^T$, this system has the solution $\mathbf{x}(t) = [e^{-t}, 0]^T$ that is bounded for all $t \geq 0$. However, this does not mean that the equilibrium $\mathbf{x}_e(t) \equiv 0$ is stable. For linear systems with constant coefficients, stability means that the solution $\mathbf{x}(t)$ remains bounded for all time and for all initial conditions, but not just for some specific initial condition. If we can find at least one initial condition that causes one of the states to approach infinity with time, then the equilibrium is unstable. For the above system, we can choose, for example, $\mathbf{x}(0) = [1, 1]^T$. In this case $\mathbf{x}(t) = [e^{-t}, e^{2t}]^T$ is unbounded, which proves that the equilibrium $\mathbf{x}_e(t) \equiv 0$ is unstable.

2. Consider the linear system $\dot{\mathbf{x}} = A\mathbf{x}$ with

$$A = \begin{bmatrix} -0.1 & -1 \\ 1 & -0.1 \end{bmatrix}.$$

The eigenvalues of A are $-0.1 \pm i$, and, hence, the equilibrium $\mathbf{x}_e(t) \equiv 0$ is asymptotically stable. Indeed, the solution of this system is given by $\mathbf{x}(t) = e^{At}\mathbf{x}(0)$, which can be written in the real form as

$$x_1(t) = e^{-0.1t}(x_1(0)\cos(t) - x_2(0)\sin(t)),$$

$$x_2(t) = e^{-0.1t}(x_1(0)\sin(t) + x_2(0)\cos(t)).$$

(See also Chapter 56.) Thus, for all initial conditions $x_1(0)$ and $x_2(0)$, the solution tends to zero as $t \rightarrow \infty$. The phase portrait for $x_1(0) = 1$ and $x_2(0) = 0$ is presented in Figure 55.4.

3. Consider the linear system $\dot{\mathbf{x}} = A\mathbf{x}$ with

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

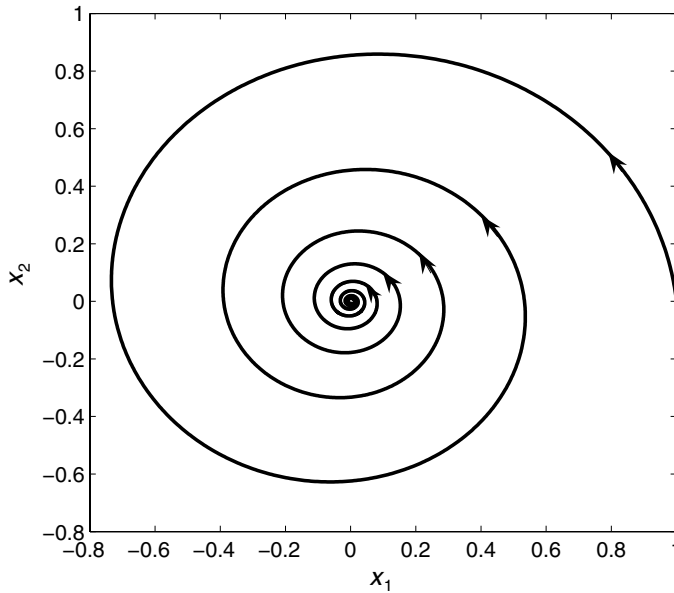


FIGURE 55.4 Asymptotic stability.

The matrix A has the eigenvalues $\pm i$. The solution of this system $\mathbf{x}(t) = e^{At}\mathbf{x}(0)$ can be written in the real form as

$$x_1(t) = x_1(0) \cos(t) - x_2(0) \sin(t),$$

$$x_2(t) = x_1(0) \sin(t) + x_2(0) \cos(t).$$

It remains bounded for all initial values $x_1(0)$ and $x_2(0)$, and, hence, the equilibrium $\mathbf{x}_e(t) \equiv 0$ is stable. The phase portrait for $x_1(0) = 1$ and $x_2(0) = 0$ is given in Figure 55.5.

4. Consider the linear system $\dot{\mathbf{x}} = A\mathbf{x}$ with

$$A = \begin{bmatrix} 0.1 & -1 \\ 1 & 0.1 \end{bmatrix}.$$

The eigenvalues of A are $0.1 \pm i$. The solution of this system in the real form is given by

$$x_1(t) = e^{0.1t}(x_1(0) \cos(t) - x_2(0) \sin(t)),$$

$$x_2(t) = e^{0.1t}(x_1(0) \sin(t) + x_2(0) \cos(t)).$$

It is unbounded for all nontrivial initial conditions. Thus, the equilibrium $\mathbf{x}_e(t) \equiv 0$ is unstable.

The phase portrait of the solution with $x_1(0) = 1$ and $x_2(0) = 0$ is shown in Figure 55.6.

5. Consider the linear system $\dot{\mathbf{x}} = A\mathbf{x}$ with

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{R}^{2 \times 2}.$$

The characteristic polynomial of the matrix A is given by $p_A(\lambda) = \lambda^2 - (a+d)\lambda + (ad-bc)$ and the eigenvalues of A have the form

$$\lambda_1 = \frac{a+d}{2} + \frac{\sqrt{(a+d)^2 - 4(ad-bc)}}{2}, \quad \lambda_2 = \frac{a+d}{2} - \frac{\sqrt{(a+d)^2 - 4(ad-bc)}}{2}.$$

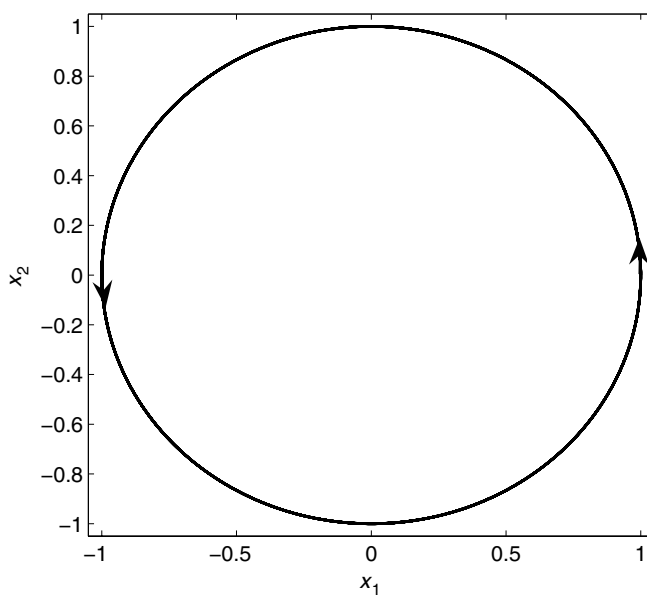


FIGURE 55.5 Stability.

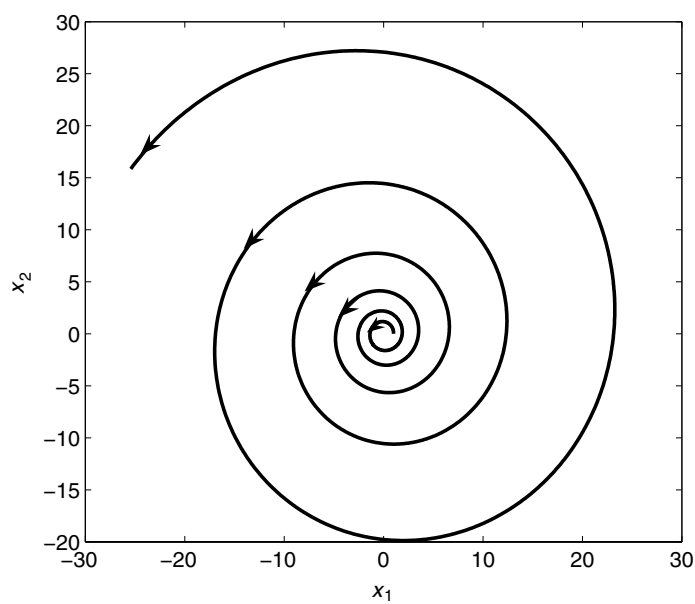


FIGURE 55.6 Instability.

We have the following cases:

$a + d < 0,$	$ad - bc > 0$	$\operatorname{Re}(\lambda_1) < 0, \operatorname{Re}(\lambda_2) < 0,$	Asymptotically stable
$a + d < 0,$	$ad - bc = 0$	$\lambda_1 = 0, \lambda_2 < 0$	Stable
$a + d < 0,$	$ad - bc < 0$	$\lambda_1 > 0, \lambda_2 < 0$	Unstable
$a + d = 0,$	$ad - bc > 0$	$\lambda_1 = i\alpha, \lambda_2 = -i\alpha, \alpha - \text{real}$	Stable
$a + d = 0,$	$ad - bc = 0$	$\lambda_1 = 0, \lambda_2 = 0$	Unstable
$a^2 + b^2 + c^2 + d^2 \neq 0$			
$a = 0, b = 0, c = 0, d = 0$		$\lambda_1 = 0, \lambda_2 = 0$	Stable
$a + d = 0,$	$ad - bc < 0$	$\lambda_1 > 0, \lambda_2 < 0$	Unstable
$a + d > 0,$	$ad - bc \leq 0$	$\lambda_1 > 0, \lambda_2 \leq 0$	Unstable
$a + d > 0,$	$ad - bc > 0$	$\operatorname{Re}(\lambda_1) > 0, \operatorname{Re}(\lambda_2) > 0$	Unstable

Applications:

1. Consider the semidiscretized heat equation; see Application 2 in Section 55.1. Let $\alpha_1 = \beta_1 = 1$ and $\alpha_2 = \beta_2 = 0$. Then $a = b = 2$ and the matrix $A_{2,2}$ has the eigenvalues

$$\lambda_j(A_{2,2}) = -\frac{4c}{h^2} \sin^2 \frac{j\pi}{2(n+1)}.$$

In this case, the equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $\dot{\mathbf{x}} = A_{2,2}\mathbf{x}$ is asymptotically stable. However, for $\alpha_1 = \beta_1 = 0$ and $\alpha_2 = \beta_2 = 1$, we have $a = b = 1$. Then the matrix $A_{1,1}$ has a simple zero eigenvalue and, hence, the equilibrium $\mathbf{x}_e(t) \equiv 0$ of $\dot{\mathbf{x}} = A_{1,1}\mathbf{x}$ is only stable.

2. Consider the mass–spring–damper model with $m > 0$, $d \geq 0$, and $k \geq 0$. The coefficient matrix of this model has eigenvalues

$$\lambda_1 = \frac{-d - \sqrt{d^2 - 4km}}{2m}, \quad \lambda_2 = \frac{-d + \sqrt{d^2 - 4km}}{2m}.$$

For $d = 0$, the equilibrium $\mathbf{x}_e(t) \equiv 0$ is unstable if $k = 0$, and it is stable if $k > 0$. For $d > 0$, the equilibrium $\mathbf{x}_e(t) \equiv 0$ is stable if $k = 0$, and it is asymptotically stable if $k > 0$.

55.5 Stability of Linear Differential-Algebraic Equations

Definitions:

The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is called **stable in the sense of Lyapunov**, or simply **stable**, if for every $\varepsilon > 0$ there exists a $\delta = \delta(\varepsilon) > 0$ such that any solution $\mathbf{x}$ of $E\dot{\mathbf{x}} = A\mathbf{x}$, $\mathbf{x}(t_0) = P_r \mathbf{x}_0$ with $\|P_r \mathbf{x}_0\|_2 < \delta$ satisfies $\|\mathbf{x}(t)\|_2 < \varepsilon$ for all $t \geq t_0$.

The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is called **asymptotically stable** if it is stable and $\lim_{t \rightarrow \infty} \mathbf{x}(t) = 0$ for every solution $\mathbf{x}$ of $E\dot{\mathbf{x}} = A\mathbf{x}$.

The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is called **unstable** if it is not stable.

The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is called **exponentially stable** if there exist constants $\alpha > 0$ and $\beta > 0$ such that the solution $\mathbf{x}$ of $E\dot{\mathbf{x}} = A\mathbf{x}$, $\mathbf{x}(t_0) = P_r \mathbf{x}_0$ satisfies $\|\mathbf{x}(t)\|_2 \leq \alpha e^{-\beta(t-t_0)} \|P_r \mathbf{x}_0\|_2$ for all $t \geq t_0$.

Facts:

1. [Dai89, pp. 68–69] If the pencil $\lambda E - A$ is regular, then the equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is stable if and only if all finite eigenvalues of the pencil $\lambda E - A$ have nonpositive real part and those with zero real part have the same algebraic and geometric multiplicities.

2. [Dai89, pp. 68–69] If the pencil $\lambda E - A$ is regular, then the equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable if and only if all finite eigenvalues of $\lambda E - A$ have negative real part.
3. [Sty02, p. 48] Let Q be a Hermitian matrix such that $\mathbf{v}^*Q\mathbf{v} > 0$ for all nonzero vectors $\mathbf{v} \in \text{range}(P_r)$. The equilibrium $\mathbf{x}_e(t) \equiv 0$ of $E\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable if the generalized Lyapunov equation $E^*XA + A^*XE = -Q$ has a Hermitian, positive semidefinite solution X .
4. [Sty02, pp. 49–52] The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable and the pencil $\lambda E - A$ is of index at most one if and only if the generalized Lyapunov equation $E^*XA + A^*XE = -E^*QE$, with Hermitian, positive definite Q has a Hermitian, positive semidefinite solution X .
5. [TMK95] The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable and the pencil $\lambda E - A$ is of index at most one if and only if the generalized Lyapunov equation

$$A^*X + Y^*A = -Q, \quad Y^*E = E^*X,$$

with Hermitian, positive definite Q has a solution (X, Y) such that E^*X is Hermitian, positive semidefinite.

6. [Sty02, pp. 52–54] The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable if and only if the projected generalized Lyapunov equation

$$E^*XA + A^*XE = -P_r^*QP_r, \quad X = P_l^*XP_l$$

has a unique Hermitian, positive semidefinite solution X for every Hermitian, positive definite matrix Q .

7. [Sty02] The equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable if and only if the projected generalized Lyapunov equation

$$EYA^* + AYE^* = -P_lQP_l^*, \quad Y = P_rYP_r^*$$

has a unique Hermitian, positive semidefinite solution Y for every Hermitian, positive definite matrix Q .

8. [Sty02, pp. 28–31] Let H be a symmetric, positive semidefinite solution of the projected generalized Lyapunov equation $E^*HA + A^*HE = -P_r^*P_r$, $H = P_l^*HP_l$ and let $\mathbf{x}$ be a solution of the initial value problem $E\dot{\mathbf{x}} = A\mathbf{x}$, $\mathbf{x}(0) = P_r\mathbf{x}_0$. Then in terms of the original data,

$$\|\mathbf{x}(t)\|_2 \leq \sqrt{\kappa(E, A)\|E\|_2\|(EP_r + A(I - P_r))^{-1}\|_2} e^{-t\|A\|_2/(\kappa(E, A)\|E\|_2)} \|P_r\mathbf{x}_0\|_2,$$

where $\kappa(E, A) = 2\|E\|_2\|A\|_2\|H\|_2$.

9. From the previous fact it follows that the equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is exponentially stable if and only if it is asymptotically stable.

Applications:

1. The finite eigenvalues of the pencil $\lambda E - A$ in the RLC electrical circuit example are given by

$$\lambda_1 = -\frac{R}{2L} - \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}, \quad \lambda_2 = -\frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}.$$

Hence, the equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable.

2. The pencil $\lambda E - A$ in the linearized pendulum example has the finite eigenvalues $\lambda_1 = -i\sqrt{g/l}$ and $\lambda_2 = i\sqrt{g/l}$. In this case the equilibrium $\mathbf{x}_e(t) \equiv 0$ of the system $E\dot{\mathbf{x}} = A\mathbf{x}$ is stable but not asymptotically stable.

Examples:

1. The generalized Lyapunov equation $E^*XA + A^*XE = -Q$ with

$$E = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

has no solution, although the finite eigenvalue of $\lambda E - A$ is negative and $\lambda E - A$ has index one.

2. The generalized Lyapunov equation $E^*XA + A^*XE = -E^*QE$ with

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad A = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

has no Hermitian, positive semidefinite solution, although the finite eigenvalue of $\lambda E - A$ is negative.

3. The generalized Lyapunov equation $A^*X + Y^*A = -Q$, $Y^*E = E^*X$ with

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

has no solution, although the finite eigenvalue of $\lambda E - A$ is negative.

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Dynamical Systems and Linear Algebra

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Linear algebra plays a key role in the theory of dynamical systems, and concepts from dynamical systems allow the study, characterization, and generalization of many objects in linear algebra, such as similarity of matrices, eigenvalues, and (generalized) eigenspaces. The most basic form of this interplay can be seen as a matrix A gives rise to a continuous time dynamical system via the linear ordinary differential equation $\dot{\mathbf{x}} = A\mathbf{x}$, or a discrete time dynamical system via iteration $\mathbf{x}_{n+1} = A\mathbf{x}_n$. The properties of the solutions are intimately related to the properties of the matrix A . Matrices also define nonlinear systems on smooth manifolds, such as the sphere $\mathbb{S}^{d-1}$ in $\mathbb{R}^d$, the Grassmann manifolds, or on classical (matrix) Lie groups. Again, the behavior of such systems is closely related to matrices and their properties. And the behavior of nonlinear systems, e.g., of differential equations $\dot{\mathbf{y}} = f(\mathbf{y})$ in $\mathbb{R}^d$ with a fixed point $\mathbf{y}_0 \in \mathbb{R}^d$, can be described locally around $\mathbf{y}_0$ via the linear differential equation $\dot{\mathbf{x}} = D_{\mathbf{y}} f(\mathbf{y}_0)\mathbf{x}$.

Since A. M. Lyapunov's thesis in 1892, it has been an intriguing problem how to construct an appropriate linear algebra for time varying systems. Note that, e.g., for stability of the solutions of $\dot{\mathbf{x}} = A(t)\mathbf{x}$, it is not sufficient that for all $t \in \mathbb{R}$ the matrices $A(t)$ have only eigenvalues with negative real part (see [Hah67], Chapter 62). Of course, Floquet theory (see [Flo83]) gives an elegant solution for the periodic case, but it is not immediately clear how to build a linear algebra around Lyapunov's "order numbers" (now called Lyapunov exponents). The multiplicative ergodic theorem of Oseledets [Ose68] resolves the issue for measurable linear systems with stationary time dependencies, and the Morse spectrum together with Selgrade's theorem [Sel75] clarifies the situation for continuous linear systems with chain transitive time dependencies.

This chapter provides a first introduction to the interplay between linear algebra and analysis/topology in continuous time. Section 56.1 recalls facts about d -dimensional linear differential equations $\dot{\mathbf{x}} = A\mathbf{x}$,

emphasizing eigenvalues and (generalized) eigenspaces. Section 56.2 studies solutions in Euclidian space $\mathbb{R}^d$ from the point of view of topological equivalence and conjugacy with related characterizations of the matrix A . Section 56.3 presents, in a fairly general set-up, the concepts of chain recurrence and Morse decompositions for dynamical systems. These ideas are then applied in section 56.4 to nonlinear systems on Grassmannian and flag manifolds induced by a single matrix A , with emphasis on characterizations of the matrix A from this point of view. Section 56.5 introduces linear skew product flows as a way to model time varying linear systems $\dot{\mathbf{x}} = A(t)\mathbf{x}$ with, e.g., periodic, measurable ergodic, and continuous chain transitive time dependencies. The following sections 56.6, 56.7, and 56.8 develop generalizations of (real parts of) eigenvalues and eigenspaces as a starting point for a linear algebra for classes of time varying linear systems, namely periodic, random, and robust systems. (For the corresponding generalization of the imaginary parts of eigenvalues see, e.g., [Arn98] for the measurable ergodic case and [CFJ06] for the continuous, chain transitive case.) Section 56.9 introduces some basic ideas to study genuinely nonlinear systems via linearization, emphasizing invariant manifolds and Grobman–Hartman-type results that compare nonlinear behavior locally to the behavior of associated linear systems.

Notation:

In this chapter, the set of $d \times d$ real matrices is denoted by $gl(d, \mathbb{R})$ rather than $\mathbb{R}^{d \times d}$.

56.1 Linear Differential Equations

Linear differential equations can be solved explicitly if one knows the eigenvalues and a basis of eigenvectors (and generalized eigenvectors, if necessary). The key idea is that of the Jordan form of a matrix. The real parts of the eigenvectors determine the exponential behavior of the solutions, described by the Lyapunov exponents and the corresponding Lyapunov subspaces.

For information on matrix functions, including the matrix exponential, see Chapter 11. For information on the Jordan canonical form see Chapter 6. Systems of first order linear differential equations are also discussed in Chapter 55.

Definitions:

For a matrix $A \in gl(d, \mathbb{R})$, the **exponential** $e^A \in GL(d, \mathbb{R})$ is defined by $e^A = I + \sum_{n=1}^{\infty} \frac{1}{n!} A^n \in GL(d, \mathbb{R})$, where $I \in gl(d, \mathbb{R})$ is the identity matrix.

A **linear differential equation** (with constant coefficients) is given by a matrix $A \in gl(d, \mathbb{R})$ via $\dot{\mathbf{x}}(t) = A\mathbf{x}(t)$, where $\dot{\mathbf{x}}$ denotes differentiation with respect to t . Any function $\mathbf{x} : \mathbb{R} \rightarrow \mathbb{R}^d$ such that $\dot{\mathbf{x}}(t) = A\mathbf{x}(t)$ for all $t \in \mathbb{R}$ is called a solution of $\dot{\mathbf{x}} = A\mathbf{x}$.

The **initial value problem** for a linear differential equation $\dot{\mathbf{x}} = A\mathbf{x}$ consists in finding, for a given **initial value** $\mathbf{x}_0 \in \mathbb{R}^d$, a solution $\mathbf{x}(\cdot, \mathbf{x}_0)$ that satisfies $\mathbf{x}(0, \mathbf{x}_0) = \mathbf{x}_0$.

The distinct (complex) eigenvalues of $A \in gl(d, \mathbb{R})$ will be denoted $\mu_1, \dots, \mu_r$. (For definitions and more information about eigenvalues, eigenvectors, and eigenspaces, see Section 4.3. For information about generalized eigenspaces, see Chapter 6.) The real version of the generalized eigenspace is denoted by $E(A, \mu_k) \subset \mathbb{R}^d$ or simply E_k for $k = 1, \dots, r \leq d$.

The **real Jordan form** of a matrix $A \in gl(d, \mathbb{R})$ is denoted by $J_A^{\mathbb{R}}$. Note that for any matrix A there is a matrix $T \in GL(d, \mathbb{R})$ such that $A = T^{-1} J_A^{\mathbb{R}} T$.

Let $\mathbf{x}(\cdot, \mathbf{x}_0)$ be a solution of the linear differential equation $\dot{\mathbf{x}} = A\mathbf{x}$. Its **Lyapunov exponent** for $\mathbf{x}_0 \neq 0$ is defined as $\lambda(\mathbf{x}_0) = \limsup_{t \rightarrow \infty} \frac{1}{t} \log \|\mathbf{x}(t, \mathbf{x}_0)\|$, where $\log$ denotes the natural logarithm and $\|\cdot\|$ is any norm in $\mathbb{R}^d$.

Let $\mu_k = \lambda_k + i\nu_k$, $k = 1, \dots, r$, be the distinct eigenvalues of $A \in gl(d, \mathbb{R})$. We order the distinct real parts of the eigenvalues as $\lambda_1 < \dots < \lambda_l$, $1 \leq l \leq r \leq d$, and define the **Lyapunov space** of λ_j as $L(\lambda_j) = \bigoplus E_k$, where the direct sum is taken over all generalized real eigenspaces associated to eigenvalues with real part equal to λ_j . Note that $\bigoplus_{j=1}^l L(\lambda_j) = \mathbb{R}^d$.

The **stable, center, and unstable subspaces** associated with the matrix $A \in gl(d, \mathbb{R})$ are defined as $L^- = \bigoplus \{L(\lambda_j), \lambda_j < 0\}$, $L^0 = \bigoplus \{L(\lambda_j), \lambda_j = 0\}$, and $L^+ = \bigoplus \{L(\lambda_j), \lambda_j > 0\}$, respectively.

The zero solution $\mathbf{x}(t, 0) \equiv 0$ is called **exponentially stable** if there exists a neighborhood $U(0)$ and positive constants $a, b > 0$ such that $\mathbf{x}(t, \mathbf{x}_0) \leq a \|\mathbf{x}_0\| e^{-bt}$ for all $t \in \mathbb{R}$ and $\mathbf{x}_0 \in U(0)$.

Facts:

Literature: [Ama90], [HSD04].

1. For each $A \in gl(d, \mathbb{R})$ the solutions of $\dot{\mathbf{x}} = A\mathbf{x}$ form a d -dimensional vector space $sol(A) \subset C^\infty(\mathbb{R}, \mathbb{R}^d)$ over $\mathbb{R}$, where $C^\infty(\mathbb{R}, \mathbb{R}^d) = \{f : \mathbb{R} \rightarrow \mathbb{R}^d, f \text{ is infinitely often differentiable}\}$. Note that the solutions of $\dot{\mathbf{x}} = A\mathbf{x}$ are even real analytic.
2. For each initial value problem given by $A \in gl(d, \mathbb{R})$ and $\mathbf{x}_0 \in \mathbb{R}^d$, the solution $\mathbf{x}(\cdot, \mathbf{x}_0)$ is unique and given by $\mathbf{x}(t, \mathbf{x}_0) = e^{At}\mathbf{x}_0$.
3. Let $\mathbf{v}_1, \dots, \mathbf{v}_d \in \mathbb{R}^d$ be a basis of $\mathbb{R}^d$. Then the functions $\mathbf{x}(\cdot, \mathbf{v}_1), \dots, \mathbf{x}(\cdot, \mathbf{v}_d)$ form a basis of the solution space $sol(A)$. The matrix function $X(\cdot) := [\mathbf{x}(\cdot, \mathbf{v}_1), \dots, \mathbf{x}(\cdot, \mathbf{v}_d)]$ is called a fundamental matrix of $\dot{\mathbf{x}} = A\mathbf{x}$, and it satisfies $\dot{X}(t) = AX(t)$.
4. Let $A \in gl(d, \mathbb{R})$ with distinct eigenvalues $\mu_1, \dots, \mu_r \in \mathbb{C}$ and corresponding multiplicities $n_k = \alpha(\mu_k)$, $k = 1, \dots, r$. If E_k are the corresponding generalized real eigenspaces, then $\dim E_k = n_k$ and $\bigoplus_{k=1}^r E_k = \mathbb{R}^d$, i.e., every matrix has a set of generalized real eigenvectors that form a basis of $\mathbb{R}^d$.
5. If $A = T^{-1}J_A^{\mathbb{R}}T$, then $e^{At} = T^{-1}e^{J_A^{\mathbb{R}}t}T$, i.e., for the computation of exponentials of matrices it is sufficient to know the exponentials of Jordan form matrices.
6. Let $\mathbf{v}_1, \dots, \mathbf{v}_d$ be a basis of generalized real eigenvectors of A . If $\mathbf{x}_0 = \sum_{i=1}^d \alpha_i \mathbf{v}_i$, then $\mathbf{x}(t, \mathbf{x}_0) = \sum_{i=1}^d \alpha_i \mathbf{x}(t, \mathbf{v}_i)$ for all $t \in \mathbb{R}$. This reduces the computation of solutions to $\dot{\mathbf{x}} = A\mathbf{x}$ to the computation of solutions for Jordan blocks; see the examples below or [HSD04, Chap. 5] for a discussion of this topic.
7. Each generalized real eigenspace E_k is invariant for the linear differential equation $\dot{\mathbf{x}} = A\mathbf{x}$, i.e., for $\mathbf{x}_0 \in E_k$ it holds that $\mathbf{x}(t, \mathbf{x}_0) \in E_k$ for all $t \in \mathbb{R}$.
8. The Lyapunov exponent $\lambda(\mathbf{x}_0)$ of a solution $\mathbf{x}(\cdot, \mathbf{x}_0)$ (with $\mathbf{x}_0 \neq 0$) satisfies $\lambda(\mathbf{x}_0) = \lim_{t \rightarrow \pm\infty} \frac{1}{t} \log \|\mathbf{x}(t, \mathbf{x}_0)\| = \lambda_j$ if and only if $\mathbf{x}_0 \in L(\lambda_j)$. Hence, associated to a matrix $A \in gl(d, \mathbb{R})$ are exactly l Lyapunov exponents, the distinct real parts of the eigenvalues of A .
9. The following are equivalent:
 - (a) The zero solution $\mathbf{x}(t, 0) \equiv 0$ of the differential equation $\dot{\mathbf{x}} = A\mathbf{x}$ is asymptotically stable.
 - (b) The zero solution is exponentially stable
 - (c) All Lyapunov exponents are negative.
 - (d) $L^- = \mathbb{R}^d$.

Examples:

1. Let $A = \text{diag}(a_1, \dots, a_d)$ be a diagonal matrix. Then the solution of the linear differential equation $\dot{\mathbf{x}} = A\mathbf{x}$ with initial value $\mathbf{x}_0 \in \mathbb{R}^d$ is given by $\mathbf{x}(t, \mathbf{x}_0) = e^{At}\mathbf{x}_0 = \begin{bmatrix} e^{a_1 t} & & \\ & \ddots & \\ & & e^{a_d t} \end{bmatrix} \mathbf{x}_0$.
2. Let $\mathbf{e}_1 = (1, 0, \dots, 0)^T, \dots, \mathbf{e}_d = (0, 0, \dots, 1)^T$ be the standard basis of $\mathbb{R}^d$. Then $\{\mathbf{x}(\cdot, \mathbf{e}_1), \dots, \mathbf{x}(\cdot, \mathbf{e}_d)\}$ is a basis of the solution space $sol(A)$.
3. Let $A = \text{diag}(a_1, \dots, a_d)$ be a diagonal matrix. Then the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_d\}$ of $\mathbb{R}^d$ consists of eigenvectors of A .

4. Let $A \in gl(d, \mathbb{R})$ be diagonalizable, i.e., there exists a transformation matrix $T \in GL(d, \mathbb{R})$ and a diagonal matrix $D \in gl(d, \mathbb{R})$ with $A = T^{-1}DT$. Then the solution of the linear differential equation $\dot{\mathbf{x}} = A\mathbf{x}$ with initial value $\mathbf{x}_0 \in \mathbb{R}^d$ is given by $\mathbf{x}(t, \mathbf{x}_0) = T^{-1}e^{Dt}T\mathbf{x}_0$, where e^{Dt} is given in Example 1.
5. Let $B = \begin{bmatrix} \lambda & -\nu \\ \nu & \lambda \end{bmatrix}$ be the real Jordan block associated with a complex eigenvalue $\mu = \lambda + i\nu$ of the matrix $A \in gl(d, \mathbb{R})$. Let $\mathbf{y}_0 \in E(A, \mu)$, the real eigenspace of μ . Then the solution $\mathbf{y}(t, \mathbf{y}_0)$ of $\dot{\mathbf{y}} = B\mathbf{y}$ is given by $\mathbf{y}(t, \mathbf{y}_0) = e^{\lambda t} \begin{bmatrix} \cos \nu t & -\sin \nu t \\ \sin \nu t & \cos \nu t \end{bmatrix} \mathbf{y}_0$. According to Fact 6 this is also the $E(A, \mu)$ -component of the solutions of $\dot{\mathbf{x}} = J_A^{\mathbb{R}}\mathbf{x}$.
6. Let B be a Jordan block of dimension n associated with the real eigenvalue μ of a matrix $A \in gl(d, \mathbb{R})$. Then for

$$B = \begin{bmatrix} \mu & 1 & & & \\ & \cdot & \cdot & & \\ & & \cdot & \cdot & \\ & & & \cdot & \cdot \\ & & & & \mu \end{bmatrix} \quad \text{one has } e^{Bt} = e^{\mu t} \begin{bmatrix} 1 & t & \frac{t^2}{2!} & \cdot & \cdot & \frac{t^{n-1}}{(n-1)!} \\ & \cdot & \cdot & \cdot & & \cdot \\ & & \cdot & \cdot & \cdot & \cdot \\ & & & \cdot & \cdot & \frac{t^2}{2!} \\ & & & & \cdot & t \\ & & & & & 1 \end{bmatrix}.$$

In other words, for $\mathbf{y}_0 = [y_1, \dots, y_n]^T \in E(A, \mu)$, the j th component of the solution of $\dot{\mathbf{y}} = B\mathbf{y}$ reads $y_j(t, \mathbf{y}_0) = e^{\mu t} \sum_{k=j}^n \frac{t^{k-j}}{(k-j)!} y_k$. According to Fact 6 this is also the $E(A, \mu)$ -component of $e^{J_A^{\mathbb{R}} t}$.

7. Let B be a real Jordan block of dimension $n = 2m$ associated with the complex eigenvalue $\mu = \lambda + i\nu$ of a matrix $A \in gl(d, \mathbb{R})$. Then with $D = \begin{bmatrix} \lambda & -\nu \\ \nu & \lambda \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, for

$$B = \begin{bmatrix} D & I & & & \\ & \cdot & \cdot & & \\ & & \cdot & \cdot & \\ & & & \cdot & \cdot \\ & & & & D \end{bmatrix} \quad \text{one has } e^{Bt} = e^{\lambda t} \begin{bmatrix} \widehat{D} & t\widehat{D} & \frac{t^2}{2!}\widehat{D} & \cdot & \cdot & \frac{t^{n-1}}{(n-1)!}\widehat{D} \\ & \cdot & \cdot & \cdot & & \cdot \\ & & \cdot & \cdot & \cdot & \cdot \\ & & & \cdot & \cdot & \frac{t^2}{2!}\widehat{D} \\ & & & & \cdot & t\widehat{D} \\ & & & & & \widehat{D} \end{bmatrix},$$

where $\widehat{D} = \begin{bmatrix} \cos \nu t & -\sin \nu t \\ \sin \nu t & \cos \nu t \end{bmatrix}$. In other words, for $\mathbf{y}_0 = [y_1, z_1, \dots, y_m, z_m]^T \in E(A, \mu)$, the j th components, $j = 1, \dots, m$, of the solution of $\dot{\mathbf{y}} = B\mathbf{y}$ read

$$y_j(t, \mathbf{y}_0) = e^{\mu t} \sum_{k=j}^m \frac{t^{k-j}}{(k-j)!} (y_k \cos \nu t - z_k \sin \nu t),$$

$$z_j(t, \mathbf{y}_0) = e^{\mu t} \sum_{k=j}^m \frac{t^{k-j}}{(k-j)!} (z_k \cos \nu t + y_k \sin \nu t).$$

According to Fact 6, this is also the $E(A, \mu)$ -component of $e^{J_A^{\mathbb{R}} t}$.

8. Using these examples and Facts 5 and 6, it is possible to compute explicitly the solutions to any linear differential equation in $\mathbb{R}^d$.
9. Recall that for any matrix A there is a matrix $T \in GL(d, \mathbb{R})$ such that $A = T^{-1}J_A^{\mathbb{R}}T$, where $J_A^{\mathbb{R}}$ is the real Jordan canonical form of A . The exponential behavior of the solutions of $\dot{\mathbf{x}} = A\mathbf{x}$ can be read off from the diagonal elements of $J_A^{\mathbb{R}}$.

56.2 Linear Dynamical Systems in $\mathbb{R}^d$

The solutions of a linear differential equation $\dot{\mathbf{x}} = A\mathbf{x}$, where $A \in gl(d, \mathbb{R})$, define a (continuous time) dynamical system, or linear flow in $\mathbb{R}^d$. The standard concepts for comparison of dynamical systems are equivalences and conjugacies that map trajectories into trajectories. For linear flows in $\mathbb{R}^d$ these concepts lead to two different classifications of matrices, depending on the smoothness of the conjugacy or equivalence.

Definitions:

The real square matrix A is **hyperbolic** if it has no eigenvalues on the imaginary axis.

A **continuous dynamical system** over the “time set” $\mathbb{R}$ with state space M , a complete metric space, is defined as a map $\Phi : \mathbb{R} \times M \longrightarrow M$ with the properties

- (i) $\Phi(0, x) = x$ for all $x \in M$,
- (ii) $\Phi(s + t, x) = \Phi(s, \Phi(t, x))$ for all $s, t \in \mathbb{R}$ and all $x \in M$,
- (iii) Φ is continuous (in both variables).

The map Φ is also called a (continuous) flow.

For each $x \in M$ the set $\{\Phi(t, x), t \in \mathbb{R}\}$ is called the **orbit (or trajectory)** of the system through x .

For each $t \in \mathbb{R}$ the **time- t map** is defined as $\varphi_t = \Phi(t, \cdot) : M \longrightarrow M$. Using time- t maps, the properties (i) and (ii) above can be restated as (i)' $\varphi_0 = id$, the identity map on M , (ii)' $\varphi_{s+t} = \varphi_s \circ \varphi_t$ for all $s, t \in \mathbb{R}$.

A **fixed point (or equilibrium)** of a dynamical system Φ is a point $x \in M$ with the property $\Phi(t, x) = x$ for all $t \in \mathbb{R}$.

An orbit $\{\Phi(t, x), t \in \mathbb{R}\}$ of a dynamical system Φ is called **periodic** if there exists $\hat{t} \in \mathbb{R}, \hat{t} > 0$ such that $\Phi(\hat{t} + s, x) = \Phi(s, x)$ for all $s \in \mathbb{R}$. The infimum of the positive $\hat{t} \in \mathbb{R}$ with this property is called the **period** of the orbit. Note that an orbit of period 0 is a fixed point.

Denote by $C^k(X, Y)$ ($k \geq 0$) the set of k -times **differentiable functions** between C^k -manifolds X and Y , with C^0 denoting continuous.

Let $\Phi, \Psi : \mathbb{R} \times M \longrightarrow M$ be two continuous dynamical systems of class C^k ($k \geq 0$), i.e., for $k \geq 1$ the state space M is at least a C^k -manifold and Φ, Ψ are C^k -maps. The flows Φ and Ψ are:

- (i) **C^k -equivalent** ($k \geq 1$) if there exists a (local) C^k -diffeomorphism $h : M \rightarrow M$ such that h takes orbits of Φ onto orbits of Ψ , **preserving the orientation** (but not necessarily parametrization by time), i.e.,
 - (a) For each $x \in M$ there is a strictly increasing and continuous parametrization map $\tau_x : \mathbb{R} \rightarrow \mathbb{R}$ such that $h(\Phi(t, x)) = \Psi(\tau_x(t), h(x))$ or, equivalently,
 - (b) For all $x \in M$ and $\delta > 0$ there exists $\varepsilon > 0$ such that for all $t \in (0, \delta)$, $h(\Phi(t, x)) = \Psi(t', h(x))$ for some $t' \in (0, \varepsilon)$.
- (ii) **C^k -conjugate** ($k \geq 1$) if there exists a (local) C^k -diffeomorphism $h : M \rightarrow M$ such that $h(\Phi(t, x)) = \Psi(t, h(x))$ for all $x \in M$ and $t \in \mathbb{R}$.

Similarly, the flows Φ and Ψ are **C^0 -equivalent** if there exists a (local) homeomorphism $h : M \rightarrow M$ satisfying the properties of (i) above, and they are **C^0 -conjugate** if there exist a (local) homeomorphism $h : M \rightarrow M$ satisfying the properties of (ii) above. Often, C^0 -equivalence is called **topological equivalence**, and C^0 -conjugacy is called **topological conjugacy** or simply **conjugacy**.

Warning: While this terminology is standard in dynamical systems, the terms *conjugate* and *equivalent* are used differently in linear algebra. *Conjugacy* as used here is related to matrix similarity (cf. Fact 6), *not* to matrix conjugacy, and *equivalence* as used here is *not* related to matrix equivalence.

Facts:

Literature: [HSD04], [Rob98].

1. If the flows Φ and Ψ are C^k -conjugate, then they are C^k -equivalent.
2. Each time- t map φ_t has an inverse $(\varphi_t)^{-1} = \varphi_{-t}$, and $\varphi_t : M \rightarrow M$ is a homeomorphism, i.e., a continuous bijective map with continuous inverse.
3. Denote the set of time- t maps again by $\Phi = \{\varphi_t, t \in \mathbb{R}\}$. A dynamical system is a group in the sense that $(\Phi, \circ)$, with $\circ$ denoting composition of maps, satisfies the group axioms, and $\varphi : (\mathbb{R}, +) \rightarrow (\Phi, \circ)$, defined by $\varphi(t) = \varphi_t$, is a group homomorphism.
4. Let M be a C^∞ -differentiable manifold and X a C^∞ -vector field on M such that the differential equation $\dot{x} = X(x)$ has unique solutions $x(t, x_0)$ for all $x_0 \in M$ and all $t \in \mathbb{R}$, with $x(0, x_0) = x_0$. Then $\Phi(t, x_0) = x(t, x_0)$ defines a dynamical system $\Phi : \mathbb{R} \times M \rightarrow M$.
5. A point $x_0 \in M$ is a fixed point of the dynamical system Φ associated with a differential equation $\dot{x} = X(x)$ as above if and only if $X(x_0) = 0$.
6. For two linear flows Φ (associated with $\dot{\mathbf{x}} = A\mathbf{x}$) and Ψ (associated with $\dot{\mathbf{x}} = B\mathbf{x}$) in $\mathbb{R}^d$, the following are equivalent:
 - Φ and Ψ are C^k -conjugate for $k \geq 1$.
 - Φ and Ψ are linearly conjugate, i.e., the conjugacy map h is a linear operator in $GL(\mathbb{R}^d)$.
 - A and B are similar, i.e., $A = TBT^{-1}$ for some $T \in GL(d, \mathbb{R})$.
7. Each of the statements in Fact 6 implies that A and B have the same eigenvalue structure and (up to a linear transformation) the same generalized real eigenspace structure. In particular, the C^k -conjugacy classes are exactly the real Jordan canonical form equivalence classes in $gl(d, \mathbb{R})$.
8. For two linear flows Φ (associated with $\dot{\mathbf{x}} = A\mathbf{x}$) and Ψ (associated with $\dot{\mathbf{x}} = B\mathbf{x}$) in $\mathbb{R}^d$, the following are equivalent:
 - Φ and Ψ are C^k -equivalent for $k \geq 1$.
 - Φ and Ψ are linearly equivalent, i.e., the equivalence map h is a linear map in $GL(\mathbb{R}^d)$.
 - $A = \alpha TBT^{-1}$ for some positive real number α and $T \in GL(d, \mathbb{R})$.
9. Each of the statements in Fact 8 implies that A and B have the same real Jordan structure and their eigenvalues differ by a positive constant. Hence, the C^k -equivalence classes are real Jordan canonical form equivalence classes modulo a positive constant.
10. The set of hyperbolic matrices is open and dense in $gl(d, \mathbb{R})$. A matrix A is hyperbolic if and only if it is structurally stable in $gl(d, \mathbb{R})$, i.e., there exists a neighborhood $U \subset gl(d, \mathbb{R})$ of A such that all $B \in U$ are topologically equivalent to A .
11. If A and B are hyperbolic, then the associated linear flows Φ and Ψ in $\mathbb{R}^d$ are C^0 -equivalent (and C^0 -conjugate) if and only if the dimensions of the stable subspaces (and, hence, the dimensions of the unstable subspaces) of A and B agree.

Examples:

1. Linear differential equations: For $A \in gl(d, \mathbb{R})$ the solutions of $\dot{\mathbf{x}} = A\mathbf{x}$ form a continuous dynamical system with time set $\mathbb{R}$ and state space $M = \mathbb{R}^d$. Here $\Phi : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is defined by $\Phi(t, \mathbf{x}_0) = \mathbf{x}(t, \mathbf{x}_0) = e^{At}\mathbf{x}_0$.
2. Fixed points of linear differential equations: A point $\mathbf{x}_0 \in \mathbb{R}^d$ is a fixed point of the dynamical system Φ associated with the linear differential equation $\dot{\mathbf{x}} = A\mathbf{x}$ if and only if $\mathbf{x}_0 \in \ker A$, the kernel of A .
3. Periodic orbits of linear differential equations: The orbit $\Phi(t, \mathbf{x}_0) := \mathbf{x}(t, \mathbf{x}_0)$, $t \in \mathbb{R}$ is periodic with period $\hat{t} > 0$ if and only if $\mathbf{x}_0$ is in the eigenspace of a nonzero complex eigenvalue with zero real part.

4. For each matrix $A \in gl(d, \mathbb{R})$ its associated linear flow in $\mathbb{R}^d$ is C^k -conjugate (and, hence, C^k -equivalent) for all $k \geq 0$ to the dynamical system associated with the Jordan form $J_A^{\mathbb{R}}$.

56.3 Chain Recurrence and Morse Decompositions of Dynamical Systems

A matrix $A \in gl(d, \mathbb{R})$ and, hence, a linear differential equation $\dot{\mathbf{x}} = A\mathbf{x}$ maps subspaces of $\mathbb{R}^d$ into subspaces of $\mathbb{R}^d$. Therefore, the matrix A also defines dynamical systems on spaces of subspaces, such as the Grassmann and the flag manifolds. These are nonlinear systems, but they can be studied via linear algebra, and vice versa; the behavior of these systems allows for the investigation of certain properties of the matrix A . The key topological concepts for the analysis of systems on compact spaces like the Grassmann and flag manifolds are chain recurrence, Morse decompositions, and attractor–repeller decompositions. This section concentrates on the first two approaches, the connection to attractor–repeller decompositions can be found, e.g., in [CK00, App. B2].

Definitions:

Given a dynamical system $\Phi : \mathbb{R} \times M \longrightarrow M$, for a subset $N \subset M$ the α -**limit set** is defined as $\alpha(N) = \{y \in M, \text{ there exist sequences } x_n \text{ in } N \text{ and } t_n \rightarrow -\infty \text{ in } \mathbb{R} \text{ with } \lim_{n \rightarrow \infty} \Phi(t_n, x_n) = y\}$, and similarly the ω -**limit set** of N is defined as $\omega(N) = \{y \in M, \text{ there exist sequences } x_n \text{ in } N \text{ and } t_n \rightarrow \infty \text{ in } \mathbb{R} \text{ with } \lim_{n \rightarrow \infty} \Phi(t_n, x_n) = y\}$.

For a flow Φ on a complete metric space M and $\varepsilon, T > 0$, an (ε, T) -**chain** from $x \in M$ to $y \in M$ is given by

$$n \in \mathbb{N}, x_0 = x, \dots, x_n = y, T_0, \dots, T_{n-1} > T$$

with

$$d(\Phi(T_i, x_i), x_{i+1}) < \varepsilon \text{ for all } i,$$

where d is the metric on M .

A set $K \subset M$ is **chain transitive** if for all $x, y \in K$ and all $\varepsilon, T > 0$ there is an (ε, T) -chain from x to y .

The **chain recurrent set** $\mathcal{CR}$ is the set of all points that are chain reachable from themselves, i.e., $\mathcal{CR} = \{x \in M, \text{ for all } \varepsilon, T > 0 \text{ there is an } (\varepsilon, T)\text{-chain from } x \text{ to } x\}$.

A set $\mathcal{M} \subset M$ is a **chain recurrent component**, if it is a maximal (with respect to set inclusion) chain transitive set. In this case $\mathcal{M}$ is a connected component of the chain recurrent set $\mathcal{CR}$.

For a flow Φ on a complete metric space M , a compact subset $K \subset M$ is called **isolated invariant**, if it is invariant and there exists a neighborhood N of K , i.e., a set N with $K \subset \text{int } N$, such that $\Phi(t, x) \in N$ for all $t \in \mathbb{R}$ implies $x \in K$.

A **Morse decomposition** of a flow Φ on a complete metric space M is a finite collection $\{\mathcal{M}_i, i = 1, \dots, l\}$ of nonvoid, pairwise disjoint, and isolated compact invariant sets such that

$$(i) \text{ For all } x \in M, \omega(x), \alpha(x) \subset \bigcup_{i=1}^l \mathcal{M}_i; \text{ and}$$

$$(ii) \text{ Suppose there are } \mathcal{M}_{j_0}, \mathcal{M}_{j_1}, \dots, \mathcal{M}_{j_n} \text{ and } x_1, \dots, x_n \in M \setminus \bigcup_{i=1}^l \mathcal{M}_i \text{ with } \alpha(x_i) \subset \mathcal{M}_{j_{i-1}} \text{ and } \omega(x_i) \subset \mathcal{M}_{j_i} \text{ for } i = 1, \dots, n; \text{ then } \mathcal{M}_{j_0} \neq \mathcal{M}_{j_n}.$$

The elements of a Morse decomposition are called **Morse sets**.

A Morse decomposition $\{\mathcal{M}_i, i = 1, \dots, l\}$ is **finer** than another decomposition $\{\mathcal{N}_j, j = 1, \dots, n\}$, if for all $\mathcal{M}_i$ there exists an index $j \in \{1, \dots, n\}$ such that $\mathcal{M}_i \subset \mathcal{N}_j$.

Facts:

Literature: [Rob98], [CK00], [ACK05].

1. For a Morse decomposition $\{\mathcal{M}_i, i = 1, \dots, l\}$ the relation $\mathcal{M}_i < \mathcal{M}_j$, given by $\alpha(x) \subset \mathcal{M}_i$ and $\omega(x) \subset \mathcal{M}_j$ for some $x \in M \setminus \bigcup_{i=1}^l \mathcal{M}_i$, induces an order.
2. Let $\Phi, \Psi : \mathbb{R} \times M \longrightarrow M$ be two dynamical systems on a state space M and let $h : M \rightarrow M$ be a topological equivalence for Φ and Ψ . Then
 - (i) The point $p \in M$ is a fixed point of Φ if and only if $h(p)$ is a fixed point of Ψ ;
 - (ii) The orbit $\Phi(\cdot, p)$ is closed if and only if $\Psi(\cdot, h(p))$ is closed;
 - (iii) If $K \subset M$ is an α -(or ω -) limit set of Φ from $p \in M$, then $h[K]$ is an α -(or ω -) limit set of Ψ from $h(p) \in M$.
 - (iv) Given, in addition, two dynamical systems $\Theta_{1,2} : \mathbb{R} \times N \longrightarrow N$, if $h : M \rightarrow M$ is a topological conjugacy for the flows Φ and Ψ on M , and $g : N \rightarrow N$ is a topological conjugacy for Θ_1 and Θ_2 on N , then the product flows $\Phi \times \Theta_1$ and $\Psi \times \Theta_2$ on $M \times N$ are topologically conjugate via $h \times g : M \times N \longrightarrow M \times N$. This result is, in general, not true for topological equivalence.
3. Topological equivalences (and conjugacies) on a compact metric space M map chain transitive sets onto chain transitive sets.
4. Topological equivalences map invariant sets onto invariant sets, and minimal closed invariant sets onto minimal closed invariant sets.
5. Topological equivalences map Morse decompositions onto Morse decompositions.

Examples:

1. Dynamical systems in $\mathbb{R}^1$: Any limit set $\alpha(x)$ and $\omega(x)$ from a single point x of a dynamical system in $\mathbb{R}^1$ consists of a single fixed point. The chain recurrent components (and the finest Morse decomposition) consist of single fixed points or intervals of fixed points. Any Morse set consists of fixed points and intervals between them.
2. Dynamical systems in $\mathbb{R}^2$: A nonempty, compact limit set of a dynamical system in $\mathbb{R}^2$, which contains no fixed points, is a closed, i.e., a periodic orbit (Poincaré–Bendixson). Any nonempty, compact limit set of a dynamical system in $\mathbb{R}^2$ consists of fixed points, connecting orbits (such as homoclinic or heteroclinic orbits), and periodic orbits.
3. Consider the following dynamical system Φ in $\mathbb{R}^2 \setminus \{0\}$, given by a differential equation in polar form for $r > 0, \theta \in [0, 2\pi)$, and $a \neq 0$:

$$\dot{r} = 1 - r, \quad \dot{\theta} = a.$$

For each $\mathbf{x} \in \mathbb{R}^2 \setminus \{0\}$ the ω -limit set is the circle $\omega(\mathbf{x}) = \mathbb{S}^1 = \{(r, \theta), r = 1, \theta \in [0, 2\pi)\}$. The state space $\mathbb{R}^2 \setminus \{0\}$ is not compact, and α -limit sets exist only for $\mathbf{y} \in \mathbb{S}^1$, for which $\alpha(\mathbf{y}) = \mathbb{S}^1$.

4. Consider the flow Φ from the previous example and a second system Ψ , given by

$$\dot{r} = 1 - r, \quad \dot{\theta} = b$$

with $b \neq 0$. Then the flows Φ and Ψ are topologically equivalent, but not conjugate if $b \neq a$.

5. An example of a flow for which the limit sets from points are strictly contained in the chain recurrent components can be obtained as follows: Let $M = [0, 1] \times [0, 1]$. Let the flow Φ on M be defined such that all points on the boundary are fixed points, and the orbits for points $(x, y) \in (0, 1) \times (0, 1)$ are straight lines $\Phi(\cdot, (x, y)) = \{(z_1, z_2), z_1 = x, z_2 \in (0, 1)\}$ with $\lim_{t \rightarrow \pm\infty} \Phi(t, (x, y)) = (x, \pm 1)$. For this system, each point on the boundary is its own α - and ω -limit set. The α -limit sets for points in the interior $(x, y) \in (0, 1) \times (0, 1)$ are of the form $\{(x, -1)\}$, and the ω -limit sets are $\{(x, +1)\}$.

The only chain recurrent component for this system is $M = [0, 1] \times [0, 1]$, which is also the only Morse set.

56.4 Linear Systems on Grassmannian and Flag Manifolds

Definitions:

The k th **Grassmannian** $\mathbb{G}_k$ of $\mathbb{R}^d$ can be defined via the following construction: Let $F(k, d)$ be the set of k -frames in $\mathbb{R}^d$, where a k -**frame** is an ordered set of k linearly independent vectors in $\mathbb{R}^d$. Two k -frames $X = [\mathbf{x}_1, \dots, \mathbf{x}_k]$ and $Y = [\mathbf{y}_1, \dots, \mathbf{y}_k]$ are said to be equivalent, $X \sim Y$, if there exists $T \in GL(k, \mathbb{R})$ with $X^T = TY^T$, where X and Y are interpreted as $d \times k$ matrices. The quotient space $\mathbb{G}_k = F(k, d)/\sim$ is a compact, $k(d - k)$ -dimensional differentiable manifold. For $k = 1$, we obtain the projective space $\mathbb{P}^{d-1} = \mathbb{G}_1$ in $\mathbb{R}^d$.

The k th **flag** of $\mathbb{R}^d$ is given by the following k -sequences of subspace inclusions,

$$\mathbb{F}_k = \{F_k = (V_1, \dots, V_k), V_i \subset V_{i+1} \text{ and } \dim V_i = i \text{ for all } i\}.$$

For $k = d$, this is the **complete flag** $\mathbb{F} = \mathbb{F}_d$.

Each matrix $A \in gl(d, \mathbb{R})$ defines a map on the subspaces of $\mathbb{R}^d$ as follows: Let $V = \text{Span}(\{\mathbf{x}_1, \dots, \mathbf{x}_k\})$. Then $AV = \text{Span}(\{A\mathbf{x}_1, \dots, A\mathbf{x}_k\})$.

Denote by $\mathbb{G}_k\Phi$ and $\mathbb{F}_k\Phi$ the induced flows on the Grassmannians and the flags, respectively.

Facts:

Literature: [Rob98], [CK00], [ACK05].

1. Let $\mathbb{P}\Phi$ be the projection onto $\mathbb{P}^{d-1}$ of a linear flow $\Phi(t, x) = e^{At}x$. Then $\mathbb{P}\Phi$ has l chain recurrent components $\{\mathcal{M}_1, \dots, \mathcal{M}_l\}$, where l is the number of different Lyapunov exponents (i.e., of different real parts of eigenvalues) of A . For each Lyapunov exponent λ_i , $\mathcal{M}_i = \mathbb{P}L_i$, the projection of the i th Lyapunov space onto $\mathbb{P}^{d-1}$. Furthermore $\{\mathcal{M}_1, \dots, \mathcal{M}_l\}$ defines the finest Morse decomposition of $\mathbb{P}\Phi$ and $\mathcal{M}_i < \mathcal{M}_j$ if and only if $\lambda_i < \lambda_j$.
2. For $A, B \in gl(d, \mathbb{R})$, let $\mathbb{P}\Phi$ and $\mathbb{P}\Psi$ be the associated flows on $\mathbb{P}^{d-1}$ and suppose that there is a topological equivalence h of $\mathbb{P}\Phi$ and $\mathbb{P}\Psi$. Then the chain recurrent components $\mathcal{N}_1, \dots, \mathcal{N}_n$ of $\mathbb{P}\Psi$ are of the form $\mathcal{N}_i = h[\mathcal{M}_i]$, where $\mathcal{M}_i$ is a chain recurrent component of $\mathbb{P}\Phi$. In particular, the number of chain recurrent components of $\mathbb{P}\Phi$ and $\mathbb{P}\Psi$ agree, and h maps the order on $\{\mathcal{M}_1, \dots, \mathcal{M}_l\}$ onto the order on $\{\mathcal{N}_1, \dots, \mathcal{N}_l\}$.
3. For $A, B \in gl(d, \mathbb{R})$ let $\mathbb{P}\Phi$ and $\mathbb{P}\Psi$ be the associated flows on $\mathbb{P}^{d-1}$ and suppose that there is a topological equivalence h of $\mathbb{P}\Phi$ and $\mathbb{P}\Psi$. Then the projected subspaces corresponding to real Jordan blocks of A are mapped onto projected subspaces corresponding to real Jordan blocks of B preserving the dimensions. Furthermore, h maps projected eigenspaces corresponding to real eigenvalues and to pairs of complex eigenvalues onto projected eigenspaces of the same type. This result shows that while C^0 -equivalence of projected linear flows on $\mathbb{P}^{d-1}$ determines the number l of distinct Lyapunov exponents, it also characterizes the Jordan structure within each Lyapunov space (but, obviously, not the size of the Lyapunov exponents nor their sign). It imposes very restrictive conditions on the eigenvalues and the Jordan structure. Therefore, C^0 -equivalences are not a useful tool to characterize l . The requirement of mapping orbits into orbits is too strong. A weakening leads to the following characterization.
4. Two matrices A and B in $gl(d, \mathbb{R})$ have the same vector of the dimensions d_i of the Lyapunov spaces (in the natural order of their Lyapunov exponents) if and only if there exist a homeomorphism $h : \mathbb{P}^{d-1} \rightarrow \mathbb{P}^{d-1}$ that maps the finest Morse decomposition of $\mathbb{P}\Phi$ onto the finest Morse decomposition of $\mathbb{P}\Psi$, i.e., h maps Morse sets onto Morse sets and preserves their orders.

5. Let $A \in gl(d, \mathbb{R})$ with associated flows Φ on $\mathbb{R}^d$ and $\mathbb{F}_k \Phi$ on the k -flag.
- (i) For every $k \in \{1, \dots, d\}$ there exists a unique finest Morse decomposition $\{ {}_k \mathcal{M}_{i_j} \}$ of $\mathbb{F}_k \Phi$, where $i_j \in \{1, \dots, d\}^k$ is a multi-index, and the number of chain transitive components in $\mathbb{F}_k$ is bounded by $\frac{d!}{(d-k)!}$.
 - (ii) Let $\mathcal{M}_i$ with $i \in \{1, \dots, d\}^k$ be a chain recurrent component in $\mathbb{F}_{k-1}$. Consider the $(d-k+1)$ -dimensional vector bundle $\pi : \mathcal{W}(\mathcal{M}_i) \rightarrow \mathcal{M}_i$ with fibers

$$\mathcal{W}(\mathcal{M}_i)_{F_{k-1}} = \mathbb{R}^d / V_{k-1} \text{ for } F_k = (V_1, \dots, V_{k-1}) \in \mathcal{M}_i \subset \mathbb{F}_{k-1}.$$

Then every chain recurrent component $\mathbb{P} \mathcal{M}_{i_j}$, $j = 1, \dots, k_i \leq d - k + 1$, of the projective bundle $\mathbb{P} \mathcal{W}(\mathcal{M}_i)$ determines a chain recurrent component ${}_k \mathcal{M}_{i_j}$ on $\mathbb{F}_k$ via

$${}_k \mathcal{M}_{i_j} = \{ F_k = (F_{k-1}, V_k) \in \mathbb{F}_k : F_{k-1} \in \mathcal{M}_i \text{ and } \mathbb{P}(V_k / V_{k-1}) \in \mathbb{P} \mathcal{M}_{i_j} \}.$$

Every chain recurrent component in $\mathbb{F}_k$ is of this form; this determines the multiindex i_j inductively for $k = 2, \dots, d$.

6. On every Grassmannian $\mathbb{G}_i$ there exists a finest Morse decomposition of the dynamical system $\mathbb{G}_i \Phi$. Its Morse sets are given by the projection of the chain recurrent components from the complete flag $\mathbb{F}$.
7. Let $A \in gl(d, \mathbb{R})$ be a matrix with flow Φ on $\mathbb{R}^d$. Let L_i , $i = 1, \dots, l$, be the Lyapunov spaces of A , i.e., their projections $\mathbb{P} L_i = \mathcal{M}_i$ are the finest Morse decomposition of $\mathbb{P} \Phi$ on the projective space. For $k = 1, \dots, d$ define the index set

$$I(k) = \{(k_1, \dots, k_m) : k_1 + \dots + k_m = k \text{ and } 0 \leq k_i \leq d_i = \dim L_i\}.$$

Then the finest Morse decomposition on the Grassmannian $\mathbb{G}_k$ is given by the sets

$$\mathcal{N}_{k_1, \dots, k_m}^k = \mathbb{G}_{k_1} L_1 \oplus \dots \oplus \mathbb{G}_{k_m} L_m, \quad (k_1, \dots, k_m) \in I(k).$$

8. For two matrices $A, B \in gl(d, \mathbb{R})$ the vector of the dimensions d_i of the Lyapunov spaces (in the natural order of their Lyapunov exponents) are identical if and only if certain graphs defined on the Grassmannians are isomorphic; see [ACK05].

Examples:

1. For $A \in gl(d, \mathbb{R})$ let Φ be its linear flow in $\mathbb{R}^d$. The flow Φ projects onto a flow $\mathbb{P} \Phi$ on $\mathbb{P}^{d-1}$, given by the differential equation

$$\dot{s} = h(s, A) = (A - s^T A s / I) s, \text{ with } s \in \mathbb{P}^{d-1}.$$

Consider the matrices

$$A = \text{diag}(-1, -1, 1) \text{ and } B = \text{diag}(-1, 1, 1).$$

We obtain the following structure for the finest Morse decompositions on the Grassmannians for A :

$$\mathbb{G}_1: \quad \mathcal{M}_1 = \{\text{Span}(\mathbf{e}_1, \mathbf{e}_2)\} \text{ and } \mathcal{M}_3 = \{\text{Span}(\mathbf{e}_3)\}$$

$$\mathbb{G}_2: \quad \mathcal{M}_{1,2} = \{\text{Span}(\mathbf{e}_1, \mathbf{e}_2)\} \text{ and } \mathcal{M}_{1,3} = \{\{\text{Span}(\mathbf{x}, \mathbf{e}_3)\} : \mathbf{x} \in \text{Span}(\mathbf{e}_1, \mathbf{e}_2)\}$$

$$\mathbb{G}_3: \quad \mathcal{M}_{1,2,3} = \{\text{Span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)\}$$

and for B we have

$$\mathbb{G}_1: \quad \mathcal{N}_1 = \{\text{Span}(\mathbf{e}_1)\} \text{ and } \mathcal{N}_2 = \{\text{Span}(\mathbf{e}_2, \mathbf{e}_3)\}$$

$$\mathbb{G}_2: \quad \mathcal{N}_{1,2} = \{\text{Span}(\mathbf{e}_1, \mathbf{x}) : \mathbf{x} \in \text{Span}(\mathbf{e}_2, \mathbf{e}_3)\} \text{ and } \mathcal{N}_{2,3} = \{\text{Span}(\mathbf{e}_2, \mathbf{e}_3)\}$$

$$\mathbb{G}_3: \quad \mathcal{N}_{1,2,3} = \{\text{Span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)\}.$$

On the other hand, the Morse sets in the full flag are given for A and B by

$$\begin{bmatrix} \mathcal{M}_{1,2,3} \\ \mathcal{M}_{1,2} \\ \mathcal{M}_1 \end{bmatrix} \leq \begin{bmatrix} \mathcal{M}_{1,2,3} \\ \mathcal{M}_{1,3} \\ \mathcal{M}_1 \end{bmatrix} \leq \begin{bmatrix} \mathcal{M}_{1,2,3} \\ \mathcal{M}_{1,3} \\ \mathcal{M}_3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} \mathcal{N}_{1,2,3} \\ \mathcal{N}_{1,2} \\ \mathcal{N}_1 \end{bmatrix} \leq \begin{bmatrix} \mathcal{N}_{1,2,3} \\ \mathcal{N}_{1,2} \\ \mathcal{N}_2 \end{bmatrix} \leq \begin{bmatrix} \mathcal{N}_{1,2,3} \\ \mathcal{N}_{2,3} \\ \mathcal{N}_2 \end{bmatrix},$$

respectively. Thus, in the full flag, the numbers and the orders of the Morse sets coincide, while on the Grassmannians (together with the projection relations between different Grassmannians) one can distinguish also the dimensions of the corresponding Lyapunov spaces. (See [ACK05] for a precise statement.)

56.5 Linear Skew Product Flows

Developing a linear algebra for time varying systems $\dot{\mathbf{x}} = A(t)\mathbf{x}$ means defining appropriate concepts to generalize eigenvalues, linear eigenspaces and their dimensions, and certain normal forms that characterize the behavior of the solutions of a time varying system and that reduce to the constant matrix case if $A(t) \equiv A \in gl(d, \mathbb{R})$. The eigenvalues and eigenspaces of the family $\{A(t), t \in \mathbb{R}\}$ do not provide an appropriate generalization; see, e.g., [Hah67], Chapter 62. For certain classes of time varying systems it turns out that the Lyapunov exponents and Lyapunov spaces introduced in section 56.1 capture the key properties of (real parts of) eigenvalues and of the associated subspace decomposition of $\mathbb{R}^d$. These systems are linear skew product flows for which the base is a (nonlinear) system θ_t that enters into the linear dynamics of a differential equation in the form $\dot{\mathbf{x}} = A(\theta_t)\mathbf{x}$. Examples for this type of systems include periodic and almost periodic differential equations, random differential equations, systems over ergodic or chain recurrent bases, linear robust systems, and bilinear control systems. This section concentrates on periodic linear differential equations, random linear dynamical systems, and robust linear systems. It is written to emphasize the correspondences between the linear algebra in Section 56.1, Floquet theory, the multiplicative ergodic theorem, and the Morse spectrum and Selgrade's theorem.

Literature: [Arn98], [BK94], [CK00], [Con97], [Rob98].

Definitions:

A (continuous time) **linear skew-product flow** is a dynamical system with state space $M = \Omega \times \mathbb{R}^d$ and flow $\Phi : \mathbb{R} \times \Omega \times \mathbb{R}^d \rightarrow \Omega \times \mathbb{R}^d$, where $\Phi = (\theta, \varphi)$ is defined as follows: $\theta : \mathbb{R} \times \Omega \rightarrow \Omega$ is a dynamical system, and $\varphi : \mathbb{R} \times \Omega \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is linear in its $\mathbb{R}^d$ -component, i.e., for each $(t, \omega) \in \mathbb{R} \times \Omega$ the map $\varphi(t, \omega, \cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is linear. Skew-product flows are called measurable (continuous, differentiable) if $\Omega = (\theta, \varphi)$ is a measurable space (topological space, differentiable manifold) and Φ is measurable (continuous, differentiable). For the time- t maps, the notation $\theta_t = \theta(t, \cdot) : \Omega \rightarrow \Omega$ is used again.

Note that the **base component** $\theta : \mathbb{R} \times \Omega \rightarrow \Omega$ is a dynamical system itself, while the **skew-component** φ is not a dynamical system. The skew-component φ is often called a **co-cycle** over θ .

Let $\Phi : \mathbb{R} \times \Omega \times \mathbb{R}^d \rightarrow \Omega \times \mathbb{R}^d$ be a linear skew-product flow. For $\mathbf{x}_0 \in \mathbb{R}^d$, $\mathbf{x}_0 \neq 0$, the **Lyapunov exponent** is defined as $\lambda(\mathbf{x}_0, \omega) = \limsup_{t \rightarrow \infty} \frac{1}{t} \log \|\varphi(t, \omega, \mathbf{x}_0)\|$, where $\log$ denotes the natural logarithm and $\|\cdot\|$ is any norm in $\mathbb{R}^d$.

Examples:

1. Time varying linear differential equations: Let $A : \mathbb{R} \rightarrow gl(d, \mathbb{R})$ be a uniformly continuous function and consider the linear differential equation $\dot{\mathbf{x}}(t) = A(t)\mathbf{x}(t)$. The solutions of this differential equation define a dynamical system via $\Phi : \mathbb{R} \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R} \times \mathbb{R}^d$, where $\theta : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is given by $\theta(t, \tau) = t + \tau$, and $\varphi : \mathbb{R} \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is defined as $\varphi(t, \tau, \mathbf{x}_0) = X(t + \tau, \tau)\mathbf{x}_0$. Here $X(t, \tau)$ is a fundamental matrix of the differential equation $\dot{X}(t) = A(t)X(t)$ in $gl(d, \mathbb{R})$. Note that for $\varphi(t, \tau, \cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $t \in \mathbb{R}$, we have $\varphi(t + s, \tau) = \varphi(t, \theta(s, \tau)) \circ \varphi(s, \tau)$ and, hence, the

solutions of $\dot{\mathbf{x}}(t) = A(t)\mathbf{x}(t)$ themselves do not define a flow. The additional component θ “keeps track of time.”

2. **Metric dynamical systems:** Let $(\Omega, \mathcal{F}, P)$ be a probability space, i.e., a set Ω with σ -algebra $\mathcal{F}$ and probability measure P . Let $\theta : \mathbb{R} \times \Omega \rightarrow \Omega$ be a measurable flow such that the probability measure P is invariant under θ , i.e., $\theta_t P = P$ for all $t \in \mathbb{R}$, where for all measurable sets $X \in \mathcal{F}$ we define $\theta_t P(X) = P\{\theta_t^{-1}(X)\} = P(X)$. Flows of this form are often called metric dynamical systems.
3. **Random linear dynamical systems:** A random linear dynamical system is a skew-product flow $\Phi : \mathbb{R} \times \Omega \times \mathbb{R}^d \rightarrow \Omega \times \mathbb{R}^d$, where $(\Omega, \mathcal{F}, P, \theta)$ is a metric dynamical system and each $\varphi : \mathbb{R} \times \Omega \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is linear in its $\mathbb{R}^d$ -component. Examples for random linear dynamical systems are given, e.g., by linear stochastic differential equations or linear differential equations with stationary background noise; see [Arn98].
4. **Robust linear systems:** Consider a linear system with time varying perturbations of the form $\dot{\mathbf{x}} = A(u(t))\mathbf{x} := A_0\mathbf{x} + \sum_{i=1}^m u_i(t)A_i\mathbf{x}$, where $A_0, \dots, A_m \in gl(d, \mathbb{R})$, $u \in \mathcal{U} = \{u : \mathbb{R} \rightarrow U, \text{ integrable on every bounded interval}\}$, and $U \subset \mathbb{R}^m$ is compact, convex with $0 \in \text{int } U$. A robust linear system defines a linear skew-product flow via the following construction: We endow $\mathcal{U}$ with the weak*-topology of $L^\infty(\mathbb{R}, U)^*$ to make it a compact, metrizable space. The base component is defined as the shift $\theta : \mathbb{R} \times \mathcal{U} \rightarrow \mathcal{U}$, $\theta(t, u(\cdot)) = u(\cdot + t)$, and the skew-component consists of the solutions $\varphi(t, u(\cdot), \mathbf{x})$, $t \in \mathbb{R}$ of the perturbed differential equation. Then $\Phi : \mathbb{R} \times \mathcal{U} \times \mathbb{R}^d \rightarrow \mathcal{U} \times \mathbb{R}^d$, $\Phi(t, u, \mathbf{x}) = (\theta(t, u), \varphi(t, u, \mathbf{x}))$ defines a continuous linear skew-product flow. The functions u can also be considered as (open loop) controls.

56.6 Periodic Linear Differential Equations: Floquet Theory

Definitions:

A **periodic linear differential equation** $\dot{\mathbf{x}} = A(\theta_t)\mathbf{x}$ is given by a matrix function $A : \mathbb{R} \rightarrow gl(d, \mathbb{R})$ that is continuous and periodic (of period $\hat{t} > 0$). As above, the solutions define a dynamical system via $\Phi : \mathbb{R} \times \mathbb{S}^1 \times \mathbb{R}^d \rightarrow \mathbb{S}^1 \times \mathbb{R}^d$, if we identify $\mathbb{R} \bmod \hat{t}$ with the circle $\mathbb{S}^1$.

Facts:

Literature: [Ama90], [GH83], [Hah67], [Sto92], [Wig96].

1. Consider the periodic linear differential equation $\dot{\mathbf{x}} = A(\theta_t)\mathbf{x}$ with period $\hat{t} > 0$. A fundamental matrix $X(t)$ of the system is of the form $X(t) = P(t)e^{Rt}$ for $t \in \mathbb{R}$, where $P(\cdot)$ is a nonsingular, differentiable, and $\hat{t}$ -periodic matrix function and $R \in gl(d, \mathbb{C})$.
2. Let $X(\cdot)$ be a fundamental solution with $X(0) = I \in GL(d, \mathbb{R})$. The matrix $X(\hat{t}) = e^{\hat{R}\hat{t}}$ is called the monodromy matrix of the system. Note that R is, in general, not uniquely determined by X , and does not necessarily have real entries. The eigenvalues α_j , $j = 1, \dots, d$ of $X(\hat{t})$ are called the characteristic multipliers of the system, and the eigenvalues $\mu_j = \lambda_j + i\nu_j$ of R are the characteristic exponents. It holds that $\mu_j = \frac{1}{\hat{t}} \log \alpha_j + \frac{2m\pi i}{\hat{t}}$, $j = 1, \dots, d$ and $m \in \mathbb{Z}$. This determines uniquely the real parts of the characteristic exponents $\lambda_j = \text{Re } \mu_j = \log |\alpha_j|$, $j = 1, \dots, d$. The λ_j are called the Floquet exponents of the system.
3. Let $\Phi = (\theta, \varphi) : \mathbb{R} \times \mathbb{S}^1 \times \mathbb{R}^d \rightarrow \mathbb{S}^1 \times \mathbb{R}^d$ be the flow associated with a periodic linear differential equation $\dot{\mathbf{x}} = A(t)\mathbf{x}$. The system has a finite number of Lyapunov exponents λ_j , $j = 1, \dots, l \leq d$. For each exponent λ_j and each $\tau \in \mathbb{S}^1$ there exists a splitting $\mathbb{R}^d = \bigoplus_{j=1}^l L(\lambda_j, \tau)$ of $\mathbb{R}^d$ into linear subspaces with the following properties:

- (a) The subspaces $L(\lambda_j, \tau)$ have the same dimension independent of τ , i.e., for each $j = 1, \dots, l$ it holds that $\dim L(\lambda_j, \sigma) = \dim L(\lambda_j, \tau) =: d_j$ for all $\sigma, \tau \in \mathbb{S}^1$.

- (b) The subspaces $L(\lambda_j, \tau)$ are invariant under the flow Φ , i.e., for each $j = 1, \dots, l$ it holds that $\varphi(t, \tau)L(\lambda_j, \tau) = L(\lambda_j, \theta(t, \tau)) = L(\lambda_j, t + \tau)$ for all $t \in \mathbb{R}$ and $\tau \in \mathbb{S}^1$.
- (c) $\lambda(\mathbf{x}, \tau) = \lim_{t \rightarrow \pm\infty} \frac{1}{t} \log \|\varphi(t, \tau, \mathbf{x})\| = \lambda_j$ if and only if $\mathbf{x} \in L(\lambda_j, \tau) \setminus \{0\}$.
4. The Lyapunov exponents of the system are exactly the Floquet exponents. The linear subspaces $L(\lambda_j, \cdot)$ are called the Lyapunov spaces (or sometimes the Floquet spaces) of the periodic matrix function $A(t)$.
 5. For each $j = 1, \dots, l \leq d$ the map $L_j : \mathbb{S}^1 \rightarrow \mathbb{G}_{d_j}$ defined by $\tau \mapsto L(\lambda_j, \tau)$ is continuous.
 6. These facts show that for periodic matrix functions $A : \mathbb{R} \rightarrow gl(d, \mathbb{R})$ the Floquet exponents and Floquet spaces replace the real parts of eigenvalues and the Lyapunov spaces, concepts that are so useful in the linear algebra of (constant) matrices $A \in gl(d, \mathbb{R})$. The number of Lyapunov exponents and the dimensions of the Lyapunov spaces are constant for $\tau \in \mathbb{S}^1$, while the Lyapunov spaces themselves depend on the time parameter τ of the periodic matrix function $A(t)$, and they form periodic orbits in the Grassmannians $\mathbb{G}_{d_j}$ and in the corresponding flag.
 7. As an application of these results, consider the problem of stability of the zero solution of $\dot{\mathbf{x}}(t) = A(t)\mathbf{x}(t)$ with period $\hat{t} > 0$: The stable, center, and unstable subspaces associated with the periodic matrix function $A : \mathbb{R} \rightarrow gl(d, \mathbb{R})$ are defined as $L^-(\tau) = \bigoplus \{L(\lambda_j, \tau), \lambda_j < 0\}$, $L^0(\tau) = \bigoplus \{L(\lambda_j, \tau), \lambda_j = 0\}$, and $L^+(\tau) = \bigoplus \{L(\lambda_j, \tau), \lambda_j > 0\}$, respectively, for $\tau \in \mathbb{S}^1$. The zero solution $\mathbf{x}(t, 0) \equiv 0$ of the periodic linear differential equation $\dot{\mathbf{x}} = A(t)\mathbf{x}$ is asymptotically stable if and only if it is exponentially stable if and only if all Lyapunov exponents are negative if and only if $L^-(\tau) = \mathbb{R}^d$ for some (and hence for all) $\tau \in \mathbb{S}^1$.
 8. Another approach to the study of time-dependent linear differential equations is via transforming an equation with bounded coefficients into an equation of known type, such as equations with constant coefficients. Such transformations are known as Lyapunov transformations; see [Hah67, Secs. 61–63].

Examples:

1. Consider the $\hat{t}$ -periodic differential equation $\dot{\mathbf{x}} = A(t)\mathbf{x}$. This equation has a nontrivial $\hat{t}$ -periodic solution iff the system has a characteristic multiplier equal to 1; see Example 2.3 for the case with constant coefficients ([Ama90, Prop. 20.12]).
2. Let H be a continuous quadratic form in $2d$ variables $x_1, \dots, x_d, y_1, \dots, y_d$ and consider the Hamiltonian system

$$\dot{x}_i = \frac{\partial H}{\partial y_i}, \dot{y}_i = -\frac{\partial H}{\partial x_i}, i = 1, \dots, d.$$

Using $\mathbf{z}^T = [\mathbf{x}^T, \mathbf{y}^T]$, we can set $H(\mathbf{x}, \mathbf{y}, t) = \mathbf{z}^T A(t) \mathbf{z}$, where $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{12}^T & A_{22} \end{bmatrix}$ with A_{11} and A_{22} symmetric, and, hence, the equation takes the form

$$\dot{\mathbf{z}} = \begin{bmatrix} A_{12}^T(t) & A_{22}(t) \\ -A_{11}(t) & -A_{12}(t) \end{bmatrix} \mathbf{z} =: P(t)\mathbf{z}.$$

Note that $-P^T(t) = QP(t)Q^{-1}$ with $Q = \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix}$, where I is the $d \times d$ identity matrix. Assume that H is $\hat{t}$ -periodic, then the equation for $\mathbf{z}$ and its adjoint have the same Floquet exponents and for each exponent λ its negative $-\lambda$ is also a Floquet exponent. Hence, the fixed point $0 \in \mathbb{R}^{2d}$ cannot be exponentially stable ([Hah67, Sec. 60]).

3. Consider the periodic linear oscillator

$$\ddot{y} + q_1(t)\dot{y} + q_2(t)y = 0.$$

Using the substitution $y = z \exp(-\frac{1}{2} \int q_1(u) du)$ one obtains Hill's differential equation

$$\ddot{z} + p(t)z = 0, \quad p(t) := q_2(t) - \frac{1}{4}q_1(t)^2 - \frac{1}{2}\dot{q}_1(t).$$

Its characteristic equation is $\lambda^2 - 2a\lambda + 1 = 0$, with a still to be determined. The multipliers satisfy the relations $\alpha_1\alpha_2 = 1$ and $\alpha_1 + \alpha_2 = 2a$. The exponential stability of the system can be analyzed using the parameter a : If $a^2 > 1$, then one of the multipliers has absolute value > 1 and, hence, the system has an unbounded solution. If $a^2 = 1$, then the system has a nontrivial periodic solution according to Example 1. If $a^2 < 1$, then the system is stable. The parameter a can often be expressed in form of a power series; see [Hah67, Sec. 62] for more details. A special case of Hill's equation is the Mathieu equation

$$\ddot{z} + (\beta_1 + \beta_2 \cos 2t)z = 0,$$

with β_1, β_2 real parameters. For this equation numerically computed stability diagrams are available; see [Sto92, Secs. VI. 3 and 4].

56.7 Random Linear Dynamical Systems

Definitions:

Let $\theta : \mathbb{R} \times \Omega \longrightarrow \Omega$ be a metric dynamical system on the probability space $(\Omega, \mathcal{F}, P)$. A set $\Delta \in \mathcal{F}$ is called **P -invariant under θ** if $P[(\theta^{-1}(t, \Delta) \setminus \Delta) \cup (\Delta \setminus \theta^{-1}(t, \Delta))] = 0$ for all $t \in \mathbb{R}$. The flow θ is called **ergodic**, if each invariant set $\Delta \in \mathcal{F}$ has P -measure 0 or 1.

Facts:

Literature: [Arn98], [Con97].

1. (Oseledets Theorem, Multiplicative Ergodic Theorem) Consider a random linear dynamical system $\Phi = (\theta, \varphi) : \mathbb{R} \times \Omega \times \mathbb{R}^d \longrightarrow \Omega \times \mathbb{R}^d$ and assume

$$\sup_{0 \leq t \leq 1} \log^+ \|\varphi(t, \omega)\| \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P}) \text{ and } \sup_{0 \leq t \leq 1} \log^+ \|\varphi(t, \omega)^{-1}\| \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P}),$$

where $\|\cdot\|$ is any norm on $GL(d, \mathbb{R})$, $\mathcal{L}^1$ is the space of integrable functions, and $\log^+$ denotes the positive part of $\log$, i.e.,

$$\log^+(x) = \begin{cases} \log(x) & \text{for } \log(x) > 0 \\ 0 & \text{for } \log(x) \leq 0. \end{cases}$$

Then there exists a set $\widehat{\Omega} \subset \Omega$ of full P -measure, invariant under the flow $\theta : \mathbb{R} \times \Omega \longrightarrow \Omega$, such that for each $\omega \in \widehat{\Omega}$ there is a splitting $\mathbb{R}^d = \bigoplus_{j=1}^{l(\omega)} L_j(\omega)$ of $\mathbb{R}^d$ into linear subspaces with the following properties:

- (a) The number of subspaces is θ -invariant, i.e., $l(\theta(t, \omega)) = l(\omega)$ for all $t \in \mathbb{R}$, and the dimensions of the subspaces are θ -invariant, i.e., $\dim L_j(\theta(t, \omega)) = \dim L_j(\omega) =: d_j(\omega)$ for all $t \in \mathbb{R}$.
- (b) The subspaces are invariant under the flow Φ , i.e., $\varphi(t, \omega)L_j(\omega) \subset L_j(\theta(t, \omega))$ for all $j = 1, \dots, l(\omega)$.
- (c) There exist finitely many numbers $\lambda_1(\omega) < \dots < \lambda_{l(\omega)}(\omega)$ in $\mathbb{R}$ (with possibly $\lambda_1(\omega) = -\infty$), such that for each $\mathbf{x} \in \mathbb{R}^d \setminus \{0\}$ the Lyapunov exponent $\lambda(\mathbf{x}, \omega)$ exists as a limit and

$\lambda(\mathbf{x}, \omega) = \lim_{t \rightarrow \pm\infty} \frac{1}{t} \log \|\varphi(t, \tau, \mathbf{x})\| = \lambda_j(\omega)$ if and only if $\mathbf{x} \in L_j(\omega) \setminus \{0\}$. The subspaces $L_j(\omega)$ are called the Lyapunov (or sometimes the Oseledets) spaces of the system Φ .

2. The following maps are measurable: $l : \Omega \rightarrow \{1, \dots, d\}$ with the discrete σ -algebra, and for each $j = 1, \dots, l(\omega)$ the maps $L_j : \Omega \rightarrow \mathbb{G}_{d_j}$ with the Borel σ -algebra, $d_j : \Omega \rightarrow \{1, \dots, d\}$ with the discrete σ -algebra, and $\lambda_j : \Omega \rightarrow \mathbb{R} \cup \{-\infty\}$ with the (extended) Borel σ -algebra.
3. If the base flow $\theta : \mathbb{R} \times \Omega \rightarrow \Omega$ is ergodic, then the maps l, d_j , and λ_j are constant on $\widehat{\Omega}$, but the Lyapunov spaces $L_j(\omega)$ still depend (in a measurable way) on $\omega \in \widehat{\Omega}$.
4. As an application of these results, we consider random linear differential equations: Let $(\Gamma, \mathcal{E}, Q)$ be a probability space and $\xi : \mathbb{R} \times \Gamma \rightarrow \mathbb{R}^m$ a stochastic process with continuous trajectories, i.e., the functions $\xi(\cdot, \gamma) : \mathbb{R} \rightarrow \mathbb{R}^m$ are continuous for all $\gamma \in \Gamma$. The process ξ can be written as a measurable dynamical system in the following way: Define $\Omega = \mathcal{C}(\mathbb{R}, \mathbb{R}^m)$, the space of continuous functions from $\mathbb{R}$ to $\mathbb{R}^m$. We denote by $\mathcal{F}$ the σ -algebra on Ω generated by the cylinder sets, i.e., by sets of the form $Z = \{\omega \in \Omega, \omega(t_1) \in F_1, \dots, \omega(t_n) \in F_n, n \in \mathbb{N}, F_i \text{ Borel sets in } \mathbb{R}^m\}$. The process ξ induces a probability measure P on $(\Omega, \mathcal{F})$ via $P(Z) = Q\{\gamma \in \Gamma, \xi(t_i, \gamma) \in F_i \text{ for } i = 1, \dots, n\}$. Define the shift $\theta : \mathbb{R} \times \Omega \rightarrow \mathbb{R} \times \Omega$ as $\theta(t, \omega(\cdot)) = \omega(t + \cdot)$. Then $(\Omega, \mathcal{F}, P, \theta)$ is a measurable dynamical system. If ξ is stationary, i.e., if for all $n \in \mathbb{N}$, and $t, t_1, \dots, t_n \in \mathbb{R}$, and all Borel sets $F_1, \dots, F_n$ in $\mathbb{R}^m$, it holds that $Q\{\gamma \in \Gamma, \xi(t_i, \gamma) \in F_i \text{ for } i = 1, \dots, n\} = Q\{\gamma \in \Gamma, \xi(t_i + t, \gamma) \in F_i \text{ for } i = 1, \dots, n\}$, then the shift θ on Ω is P -invariant, and $(\Omega, \mathcal{F}, P, \theta)$ is a metric dynamical system.
5. Let $A : \Omega \rightarrow gl(d, \mathbb{R})$ be measurable with $A \in \mathcal{L}^1$. Consider the random linear differential equation $\dot{\mathbf{x}}(t) = A(\theta(t, \omega))\mathbf{x}(t)$, where $(\Omega, \mathcal{F}, P, \theta)$ is a metric dynamical system as described before. We understand the solutions of this equation to be ω -wise. Then the solutions define a random linear dynamical system. Since we assume that $A \in \mathcal{L}^1$, this system satisfies the integrability conditions of the Multiplicative Ergodic Theorem.
6. Hence, for random linear differential equations $\dot{\mathbf{x}}(t) = A(\theta(t, \omega))\mathbf{x}(t)$ the Lyapunov exponents and the associated Oseledets spaces replace the real parts of eigenvalues and the Lyapunov spaces of constant matrices $A \in gl(d, \mathbb{R})$. If the “background” process ξ is ergodic, then all the quantities in the Multiplicative Ergodic Theorem are constant, except for the Lyapunov spaces that do, in general, depend on chance.
7. The problem of stability of the zero solution of $\dot{\mathbf{x}}(t) = A(\theta(t, \omega))\mathbf{x}(t)$ can now be analyzed in analogy to the case of a constant matrix or a periodic matrix function: The stable, center, and unstable subspaces associated with the random matrix process $A(\theta(t, \omega))$ are defined as $L^-(\omega) = \bigoplus\{L_j(\omega), \lambda_j(\omega) < 0\}$, $L^0(\omega) = \bigoplus\{L_j(\omega), \lambda_j(\omega) = 0\}$, and $L^+(\omega) = \bigoplus\{L_j(\omega), \lambda_j(\omega) > 0\}$, respectively for $\omega \in \widehat{\Omega}$. We obtain the following characterization of stability: The zero solution $\mathbf{x}(t, \omega, 0) \equiv 0$ of the random linear differential equation $\dot{\mathbf{x}}(t) = A(\theta(t, \omega))\mathbf{x}(t)$ is P -almost surely exponentially stable if and only if P -almost surely all Lyapunov exponents are negative if and only if $P\{\omega \in \Omega, L^-(\omega) = \mathbb{R}^d\} = 1$.

Examples:

1. The case of constant matrices: Let $A \in gl(d, \mathbb{R})$ and consider the dynamical system $\varphi : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ generated by the solutions of the linear differential equation $\dot{\mathbf{x}} = A\mathbf{x}$. The flow φ can be considered as the skew-component of a random linear dynamical system over the base flow given by $\Omega = \{0\}$, $\mathcal{F}$ the trivial σ -algebra, P the Dirac measure at $\{0\}$, and $\theta : \mathbb{R} \times \Omega \rightarrow \Omega$ defined as the constant map $\theta(t, \omega) = \omega$ for all $t \in \mathbb{R}$. Since the flow is ergodic and satisfies the integrability condition, we can recover all the results on Lyapunov exponents and Lyapunov spaces for φ from the Multiplicative Ergodic Theorem.
2. Weak Floquet theory: Let $A : \mathbb{R} \rightarrow gl(d, \mathbb{R})$ be a continuous, periodic matrix function. Define the base flow as follows: $\Omega = \mathbb{S}^1$, $\mathcal{B}$ is the Borel σ -algebra on $\mathbb{S}^1$, P is the uniform distribution on $\mathbb{S}^1$, and θ is the shift $\theta(t, \tau) = t + \tau$. Then $(\Omega, \mathcal{F}, P, \theta)$ is an ergodic metric dynamical system. The solutions $\varphi(\cdot, \tau, \mathbf{x})$ of $\dot{\mathbf{x}} = A(t)\mathbf{x}$ define a random linear dynamical system $\Phi : \mathbb{R} \times \Omega \times \mathbb{R}^d \rightarrow \Omega \times \mathbb{R}^d$ via

$\Phi(t, \omega, \mathbf{x}) = (\theta(t, \omega), \varphi(t, \omega, \mathbf{x}))$. With this set-up, the Multiplicative Ergodic Theorem recovers the results of Floquet Theory with P -probability 1.

3. Average Lyapunov exponent: In general, Lyapunov exponents for random linear systems are difficult to compute explicitly—numerical methods are usually the way to go. In the ergodic case, the average Lyapunov exponent $\bar{\lambda} := \frac{1}{d} \sum d_j \lambda_j$ is given by $\bar{\lambda} = \frac{1}{d} \text{tr } E(A \mid \mathcal{I})$, where $A : \Omega \rightarrow \text{gl}(d, \mathbb{R})$ is the random matrix of the system, and $E(\cdot, \mathcal{I})$ is the conditional expectation of the probability measure P given the σ -algebra $\mathcal{I}$ of invariant sets on Ω . As an example, consider the linear oscillator with random restoring force

$$\ddot{y}(t) + 2\beta \dot{y}(t) + (1 + \sigma f(\theta(t, \omega)))y(t) = 0,$$

where $\beta, \sigma \in \mathbb{R}$ are positive constants and $f : \Omega \rightarrow \mathbb{R}$ is in $\mathcal{L}^1$. We assume that the background process is ergodic. Using the notation $x_1 = y$ and $x_2 = \dot{y}$ we can write the equation as

$$\dot{\mathbf{x}}(t) = A(\theta(t, \omega))\mathbf{x}(t) = \begin{bmatrix} 0 & 1 \\ -1 - \sigma f(\theta(t, \omega)) & -2\beta \end{bmatrix} \mathbf{x}(t).$$

For this system we obtain $\bar{\lambda} = -\beta$ ([Arn98, Remark 3.3.12]).

56.8 Robust Linear Systems

Definitions:

Let $\Phi : \mathbb{R} \times \mathcal{U} \times \mathbb{R}^d \rightarrow \mathcal{U} \times \mathbb{R}^d$ be a linear skew-product flow with continuous base flow $\theta : \mathbb{R} \times \mathcal{U} \rightarrow \mathcal{U}$. Throughout this section, $\mathcal{U}$ is compact and θ is chain recurrent on $\mathcal{U}$. Denote by $\mathcal{U} \times \mathbb{P}^{d-1}$ the projective bundle and recall that Φ induces a dynamical system $\mathbb{P}\Phi : \mathbb{R} \times \mathcal{U} \times \mathbb{P}^{d-1} \rightarrow \mathcal{U} \times \mathbb{P}^{d-1}$. For $\varepsilon, T > 0$ an (ε, T) -chain ζ of $\mathbb{P}\Phi$ is given by $n \in \mathbb{N}$, $T_0, \dots, T_n \geq T$, and $(u_0, p_0), \dots, (u_n, p_n) \in \mathcal{U} \times \mathbb{P}^{d-1}$ with $d(\mathbb{P}\Phi(T_i, u_i, p_i), (u_{i+1}, p_{i+1})) < \varepsilon$ for $i = 0, \dots, n-1$.

Define the **finite time exponential growth rate** of such a chain ζ (or **chain exponent**) by

$$\lambda(\zeta) = \left(\sum_{i=0}^{n-1} T_i \right)^{-1} \sum_{i=0}^{n-1} (\log \|\varphi(T_i, x_i, u_i)\| - \log \|x_i\|),$$

where $x_i \in \mathbb{P}^{-1}(p_i)$.

Let $\mathcal{M} \subset \mathcal{U} \times \mathbb{P}^{d-1}$ be a chain recurrent component of the flow $\mathbb{P}\Phi$. Define the **Morse spectrum over $\mathcal{M}$** as

$$\Sigma_{Mo}(\mathcal{M}) = \left\{ \lambda \in \mathbb{R}, \text{ there exist sequences } \varepsilon_n \rightarrow 0, T_n \rightarrow \infty \text{ and } \right. \\ \left. (\varepsilon_n, T_n)\text{-chains } \zeta_n \text{ in } \mathcal{M} \text{ such that } \lim \lambda(\zeta_n) = \lambda \right\}$$

and the **Morse spectrum of the flow** as

$$\Sigma_{Mo}(\Phi) = \left\{ \lambda \in \mathbb{R}, \text{ there exist sequences } \varepsilon_n \rightarrow 0, T_n \rightarrow \infty \text{ and } (\varepsilon_n, T_n)\text{-} \right. \\ \left. \text{chains } \zeta_n \text{ in the chain recurrent set of } \mathbb{P}\Phi \text{ such that } \lim \lambda(\zeta_n) = \lambda \right\}.$$

Define the **Lyapunov spectrum over $\mathcal{M}$** as

$$\Sigma_{Ly}(\mathcal{M}) = \{\lambda(u, x), (u, x) \in \mathcal{M}, x \neq 0\}$$

and the **Lyapunov spectrum of the flow Φ** as

$$\Sigma_{Ly}(\Phi) = \{\lambda(u, \mathbf{x}), (u, \mathbf{x}) \in \mathcal{U} \times \mathbb{R}^d, \mathbf{x} \neq 0\}.$$

Facts:

Literature: [CK00], [Gru96], [HP05].

1. The projected flow $\mathbb{P}\Phi$ has a finite number of chain-recurrent components $\mathcal{M}_1, \dots, \mathcal{M}_l, l \leq d$. These components form the finest Morse decomposition for $\mathbb{P}\Phi$, and they are linearly ordered $\mathcal{M}_1 < \dots < \mathcal{M}_l$. Their lifts $\mathbb{P}^{-1}\mathcal{M}_i \subset \mathcal{U} \times \mathbb{R}^d$ form a continuous subbundle decomposition of $\mathcal{U} \times \mathbb{R}^d = \bigoplus_{i=1}^l \mathbb{P}^{-1}\mathcal{M}_i$.
2. The Lyapunov spectrum and the Morse spectrum are defined on the Morse sets, i.e., $\Sigma_{Ly}(\Phi) = \bigcup_{i=1}^l \Sigma_{Ly}(\mathcal{M}_i)$ and $\Sigma_{Mo}(\Phi) = \bigcup_{i=1}^l \Sigma_{Mo}(\mathcal{M}_i)$.
3. For each Morse set $\mathcal{M}_i$ the Lyapunov spectrum is contained in the Morse spectrum, i.e., $\Sigma_{Ly}(\mathcal{M}_i) \subset \Sigma_{Mo}(\mathcal{M}_i)$ for $i = 1, \dots, l$.
4. For each Morse set, its Morse spectrum is a closed, bounded interval $\Sigma_{Mo}(\mathcal{M}_i) = [\kappa_i^*, \kappa_i]$, and $\kappa_i^*, \kappa_i \in \Sigma_{Ly}(\mathcal{M})$ for $i = 1, \dots, l$.
5. The intervals of the Morse spectrum are ordered according to the order of the Morse sets, i.e., $\mathcal{M}_i < \mathcal{M}_j$ is equivalent to $\kappa_i^* < \kappa_j^*$ and $\kappa_i < \kappa_j$.
6. As an application of these results, consider robust linear systems of the form $\Phi : \mathbb{R} \times \mathcal{U} \times \mathbb{R}^d \longrightarrow \mathcal{U} \times \mathbb{R}^d$, given by a perturbed linear differential equation $\dot{\mathbf{x}} = A(u(t))\mathbf{x} := A_0\mathbf{x} + \sum_{i=1}^m u_i(t)A_i\mathbf{x}$, with $A_0, \dots, A_m \in gl(d, \mathbb{R})$, $u \in \mathcal{U} = \{u : \mathbb{R} \longrightarrow U, \text{ integrable on every bounded interval}\}$ and $U \subset \mathbb{R}^m$ is compact, convex with $0 \in \text{int}U$. Explicit equations for the induced perturbed system on the projective space $\mathbb{P}^{d-1}$ can be obtained as follows: Let $\mathbb{S}^{d-1} \subset \mathbb{R}^d$ be the unit sphere embedded into $\mathbb{R}^d$. The projected system on $\mathbb{S}^{d-1}$ is given by

$$\dot{s}(t) = h(u(t), s(t)), \quad u \in \mathcal{U}, \quad s \in \mathbb{S}^{d-1}$$

where

$$h(u, s) = h_0(s) + \sum_{i=1}^m u_i h_i(s) \text{ with } h_i(s) = (A_i - s^T A_i s \cdot I) s, \quad i = 0, 1, \dots, m.$$

Define an equivalence relation on $\mathbb{S}^{d-1}$ via $s_1 \sim s_2$ if $s_1 = -s_2$, identifying opposite points. Then the projective space can be identified as $\mathbb{P}^{d-1} = \mathbb{S}^{d-1} / \sim$. Since $h(u, s) = -h(u, -s)$, the differential equation also describes the projected system on $\mathbb{P}^{d-1}$. For the Lyapunov exponents one obtains in the same way

$$\lambda(u, \mathbf{x}) = \limsup_{t \rightarrow \infty} \frac{1}{t} \log \|\mathbf{x}(t)\| = \limsup_{t \rightarrow \infty} \frac{1}{t} \int_0^t q(u(\tau), s(\tau)) d\tau$$

with

$$q(u, s) = q_0(s) + \sum_{i=1}^m u_i q_i(s) \text{ with } q_i(s) = (A_i - s^T A_i s \cdot I) s, \quad i = 0, 1, \dots, m.$$

For a constant perturbation $u(t) \equiv u \in \mathbb{R}$ for all $t \in \mathbb{R}$ the corresponding Lyapunov exponents $\lambda(u, \mathbf{x})$ of the flow Φ are the real parts of the eigenvalues of the matrix $A(u)$ and the corresponding Lyapunov spaces are contained in the subbundles $\mathbb{P}^{-1}\mathcal{M}_i$. Similarly, if a perturbation $u \in \mathcal{U}$ is periodic, the Floquet exponents of $\dot{\mathbf{x}} = A(u(\cdot))\mathbf{x}$ are part of the Lyapunov (and, hence, of the Morse) spectrum of the flow Φ , and the Floquet spaces are contained in $\mathbb{P}^{-1}\mathcal{M}_i$. The systems treated in this example can also be considered as “bilinear control systems” and studied relative to their control behavior and (exponential) stabilizability — this is the point of view taken in [CK00].

7. For robust linear systems “generically” the Lyapunov spectrum and the Morse spectrum agree see [CK00] for a precise definition of “generic” in this context.
8. Of particular interest is the upper spectral interval $\Sigma_{Mo}(\mathcal{M}_l) = [\kappa_l^*, \kappa_l]$, as it determines the robust stability of $\dot{\mathbf{x}} = A(u(t))\mathbf{x}$ (and stabilizability of the system if the set $\mathcal{U}$ is interpreted as a

set of admissible control functions; see [Gru96]). The stable, center, and unstable subbundles of $\mathcal{U} \times \mathbb{R}^d$ associated with the perturbed linear system $\dot{\mathbf{x}} = A(u(t))\mathbf{x}$ are defined as $L^- = \bigoplus \{\mathbb{P}^{-1}\mathcal{M}_j, \kappa_j < 0\}$, $L^0 = \bigoplus \{\mathbb{P}^{-1}\mathcal{M}_j, 0 \in [\kappa_j^*, \kappa_j]\}$, and $L^+ = \bigoplus \{\mathbb{P}^{-1}\mathcal{M}_j, \kappa_j^* > 0\}$, respectively. The zero solution of $\dot{\mathbf{x}} = A(u(t))\mathbf{x}$ is exponentially stable for all perturbations $u \in \mathcal{U}$ if and only if $\kappa_l < 0$ if and only if $L^- = \mathcal{U} \times \mathbb{R}^d$.

Examples:

1. In general, it is not possible to compute the Morse spectrum and the associated subbundle decompositions explicitly, even for relatively simple systems, and one has to revert to numerical algorithms; compare [CK00, App. D]. Let us consider, e.g., the linear oscillator with uncertain restoring force

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -2b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + u(t) \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

with $u(t) \in [-\rho, \rho]$ and $b > 0$. Figure 56.1 shows the spectral intervals for this system depending on $\rho \geq 0$.

2. We consider robust linear systems as described in Fact 6, with varying perturbation range by introducing the family $U^\rho = \rho U$ for $\rho \geq 0$. The resulting family of systems is

$$\dot{\mathbf{x}}^\rho = A(u^\rho(t))\mathbf{x}^\rho := A_0\mathbf{x}^\rho + \sum_{i=1}^m u_i^\rho(t)A_i\mathbf{x}^\rho,$$

with $u^\rho \in \mathcal{U}^\rho = \{u : \mathbb{R} \rightarrow U^\rho, \text{ integrable on every bounded interval}\}$. The corresponding maximal spectral value $\kappa_l(\rho)$ is continuous in ρ and we define the (asymptotic) stability radius of this family as $r = \inf\{\rho \geq 0, \text{ there exists } u_0 \in \mathcal{U}^\rho \text{ such that } \dot{\mathbf{x}}^\rho = A(u_0(t))\mathbf{x}^\rho \text{ is not exponentially stable}\}$. This stability radius is based on asymptotic stability under all time varying perturbations. Similarly one can introduce stability radii based on time invariant perturbations (with values in $\mathbb{R}^m$ or $\mathbb{C}^m$) or on quadratic Lyapunov functions ([CK00], Chapter 11 and [HP05]).

3. Linear oscillator with uncertain damping: Consider the oscillator

$$\ddot{y} + 2(b + u(t))\dot{y} + (1 + c)y = 0$$

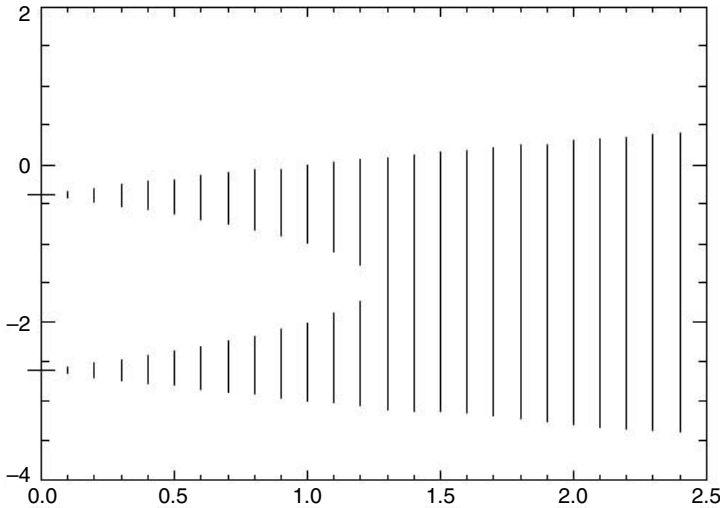


FIGURE 56.1 Spectral intervals depending on $\rho \geq 0$ for the system in Example 1.

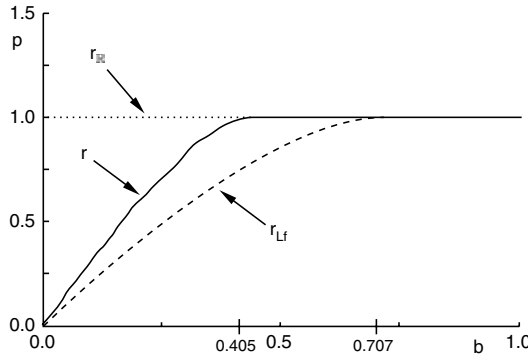


FIGURE 56.2 Stability radii for the system in Example 4.

with $u(t) \in [-\rho, \rho]$ and $c \in \mathbb{R}$. In equivalent first-order form the system reads

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1-c & -2b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + u(t) \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Clearly, the system is not exponentially stable for $c \leq -1$ with $\rho = 0$, and for $c > -1$ with $\rho \geq b$. It turns out that the stability radius for this system is

$$r(c) = \begin{cases} 0 & \text{for } c \leq -1 \\ b & \text{for } c > -1. \end{cases}$$

4. Linear oscillator with uncertain restoring force: Here we look again at a system of the form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -2b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + u(t) \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

with $u(t) \in [-\rho, \rho]$ and $b > 0$. (For $b \leq 0$ the system is unstable even for constant perturbations.) A closed form expression of the stability radius for this system is not available and one has to use numerical methods for the computation of (maximal) Lyapunov exponents (or maxima of the Morse spectrum); compare [CK00, App. D]. Figure 56.2 shows the (asymptotic) stability radius r , the stability radius under constant real perturbations $r_{\mathbb{R}}$, and the stability radius based on quadratic Lyapunov functions r_{Lf} , all in dependence on $b > 0$; see [CK00, Ex. 11.1.12].

56.9 Linearization

The local behavior of the dynamical system induced by a nonlinear differential equation can be studied via the linearization of the flow. At a fixed point of the nonlinear system the linearization is just a linear differential equation as studied in Sections 56.1 to 56.4. If the linearized system is hyperbolic, then the theorem of Hartman and Grobman states that the nonlinear flow is topologically conjugate to the linear flow. The invariant manifold theorem deals with those solutions of the nonlinear equation that are asymptotically attracted to (or repelled from) a fixed point. Basically these solutions live on manifolds that are described by nonlinear changes of coordinates of the linear stable (and unstable) subspaces.

Fact 4 below describes the simplest form of the invariant manifold theorem at a fixed point. It can be extended to include a “center manifold” (corresponding to the Lyapunov space with exponent 0). Furthermore, (local) invariant manifolds can be defined not just for the stable and unstable subspace,

but for all Lyapunov spaces; see [BK94], [CK00], and [Rob98] for the necessary techniques and precise statements.

Both the Grobman–Hartman theorem as well as the invariant manifold theorem can be extended to time varying systems, i.e., to linear skew product flows as described in Sections 56.5 to 56.8. The general situation is discussed in [BK94], the case of linearization at a periodic solution is covered in [Rob98], random dynamical systems are treated in [Arn98], and robust systems (control systems) are the topic of [CK00].

Definitions:

A (nonlinear) **differential equation** in $\mathbb{R}^d$ is of the form $\dot{\mathbf{y}} = f(\mathbf{y})$, where f is a vector field on $\mathbb{R}^d$. Assume that f is at least of class C^1 and that for all $\mathbf{y}_0 \in \mathbb{R}^d$ the solutions $\mathbf{y}(t, \mathbf{y}_0)$ of the initial value problem $\mathbf{y}(0, \mathbf{y}_0) = \mathbf{y}_0$ exist for all $t \in \mathbb{R}$.

A point $\mathbf{p} \in \mathbb{R}^d$ is a **fixed point** of the differential equation $\dot{\mathbf{y}} = f(\mathbf{y})$ if $\mathbf{y}(t, \mathbf{p}) = \mathbf{p}$ for all $t \in \mathbb{R}$.

The **linearization** of the equation $\dot{\mathbf{y}} = f(\mathbf{y})$ at a fixed point $\mathbf{p} \in \mathbb{R}^d$ is given by $\dot{\mathbf{x}} = D_{\mathbf{y}}f(\mathbf{p})\mathbf{x}$, where $D_{\mathbf{y}}f(\mathbf{p})$ is the Jacobian (matrix of partial derivatives) of f at the point $\mathbf{p}$.

A fixed point $\mathbf{p} \in \mathbb{R}^d$ of the differential equation $\dot{\mathbf{y}} = f(\mathbf{y})$ is called **hyperbolic** if $D_{\mathbf{y}}f(\mathbf{p})$ has no eigenvalues on the imaginary axis, i.e., if the matrix $D_{\mathbf{y}}f(\mathbf{p})$ is hyperbolic.

Consider a differential equation $\dot{\mathbf{y}} = f(\mathbf{y})$ in $\mathbb{R}^d$ with flow $\Phi : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$, hyperbolic fixed point $\mathbf{p}$ and neighborhood $U(\mathbf{p})$. In this situation the **local stable manifold** and the **local unstable manifold** are defined as

$$W_{loc}^s(\mathbf{p}) = \{\mathbf{q} \in U : \lim_{t \rightarrow \infty} \Phi(t, \mathbf{q}) = \mathbf{p}\} \text{ and } W_{loc}^u(\mathbf{p}) = \{\mathbf{q} \in U : \lim_{t \rightarrow -\infty} \Phi(t, \mathbf{q}) = \mathbf{p}\},$$

respectively.

The local stable (and unstable) manifolds can be extended to **global invariant manifolds** by following the trajectories, i.e.,

$$W^s(\mathbf{p}) = \bigcup_{t \geq 0} \Phi(-t, W_{loc}^s(\mathbf{p})) \text{ and } W^u(\mathbf{p}) = \bigcup_{t \geq 0} \Phi(t, W_{loc}^u(\mathbf{p})).$$

Facts:

Literature: [Arn98], [AP90], [BK94], [CK00], [Rob98].

See Facts 3 and 4 in Section 56.2 for dynamical systems induced by differential equations and their fixed points.

1. (Hartman–Gobman) Consider a differential equation $\dot{\mathbf{y}} = f(\mathbf{y})$ in $\mathbb{R}^d$ with flow $\Phi : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$. Assume that the equation has a hyperbolic fixed point $\mathbf{p}$ and denote the flow of the linearized equation $\dot{\mathbf{x}} = D_{\mathbf{y}}f(\mathbf{p})\mathbf{x}$ by $\Psi : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$. Then there exist neighborhoods $U(\mathbf{p})$ of $\mathbf{p}$ and $V(\mathbf{0})$ of the origin in $\mathbb{R}^d$, and a homeomorphism $h : U(\mathbf{p}) \rightarrow V(\mathbf{0})$ such that the flows $\Phi|_{U(\mathbf{p})}$ and $\Psi|_{V(\mathbf{0})}$ are (locally) C^0 -conjugate, i.e., $h(\Phi(t, \mathbf{y})) = \Psi(t, h(\mathbf{y}))$ for all $\mathbf{y} \in U(\mathbf{p})$ and $t \in \mathbb{R}$ as long as the solutions stay within the respective neighborhoods.
2. Consider two differential equations $\dot{\mathbf{y}} = f_i(\mathbf{y})$ in $\mathbb{R}^d$ with flows $\Phi_i : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ for $i = 1, 2$. Assume that Φ_i has a hyperbolic fixed point $\mathbf{p}_i$ and the flows are C^k -conjugate for some $k \geq 1$ in neighborhoods of the $\mathbf{p}_i$. Then $\sigma(D_{\mathbf{y}}f_1(\mathbf{p}_1)) = \sigma(D_{\mathbf{y}}f_2(\mathbf{p}_2))$, i.e., the eigenvalues of the linearizations agree; compare Facts 5 and 6 in Section 56.2 for the linear situation.
3. Consider two differential equations $\dot{\mathbf{y}} = f_i(\mathbf{y})$ in $\mathbb{R}^d$ with flows $\Phi_i : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ for $i = 1, 2$. Assume that Φ_i has a hyperbolic fixed point $\mathbf{p}_i$ and the number of negative (or positive) Lyapunov exponents of $D_{\mathbf{y}}f_i(\mathbf{p}_i)$ agrees. Then the flows Φ_i are locally C^0 -conjugate around the fixed points.
4. (Invariant Manifold Theorem) Consider a differential equation $\dot{\mathbf{y}} = f(\mathbf{y})$ in $\mathbb{R}^d$ with flow $\Phi : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$. Assume that the equation has a hyperbolic fixed point p and denote the linearized equation by $\dot{\mathbf{x}} = D_{\mathbf{y}}f(p)\mathbf{x}$.

- (i) There exists a neighborhood $U(\mathbf{p})$ in which the flow Φ has a local stable manifold $W_{loc}^s(\mathbf{p})$ and a local unstable manifold $W_{loc}^u(\mathbf{p})$.

- (ii) Denote by L^- (and L^+) the stable (and unstable, respectively) subspace of $D_{\mathbf{y}}f(\mathbf{p})$; compare the definitions in Section 56.1. The dimensions of L^- (as a linear subspace of $\mathbb{R}^d$) and of $W_{loc}^s(\mathbf{p})$ (as a topological manifold) agree, similarly for L^+ and $W_{loc}^u(\mathbf{p})$.
- (iii) The stable manifold $W_{loc}^s(\mathbf{p})$ is tangent to the stable subspace L^- at the fixed point $\mathbf{p}$, similarly for $W_{loc}^u(\mathbf{p})$ and L^+ .
5. Consider a differential equation $\dot{\mathbf{y}} = f(\mathbf{y})$ in $\mathbb{R}^d$ with flow $\Phi : \mathbb{R} \times \mathbb{R}^d \longrightarrow \mathbb{R}^d$. Assume that the equation has a hyperbolic fixed point $\mathbf{p}$. Then there exists a neighborhood $U(\mathbf{p})$ on which Φ is C^0 -equivalent to the flow of a linear differential equation of the type

$$\begin{aligned}\dot{\mathbf{x}}_s &= -\mathbf{x}_s, \mathbf{x}_s \in \mathbb{R}^{d_s}, \\ \dot{\mathbf{x}}_u &= \mathbf{x}_u, \mathbf{x}_u \in \mathbb{R}^{d_u},\end{aligned}$$

where d_s and d_u are the dimensions of the stable and the unstable subspace of $D_{\mathbf{y}}f(\mathbf{p})$, respectively, with $d_s + d_u = d$.

Examples:

1. Consider the nonlinear differential equation in $\mathbb{R}$ given by $\ddot{z} + z - z^3 = 0$, or in first-order form in $\mathbb{R}^2$

$$\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} = \begin{bmatrix} y_2 \\ -y_1 + y_1^3 \end{bmatrix} = f(\mathbf{y}).$$

The fixed points of this system are $\mathbf{p}_1 = [0, 0]^T$, $\mathbf{p}_2 = [1, 0]^T$, $\mathbf{p}_3 = [-1, 0]^T$. Computation of the linearization yields

$$D_{\mathbf{y}}f = \begin{bmatrix} 0 & 1 \\ -1 + 3y_1^2 & 0 \end{bmatrix}.$$

Hence, the fixed point $\mathbf{p}_1$ is not hyperbolic, while $\mathbf{p}_2$ and $\mathbf{p}_3$ have this property.

2. Consider the nonlinear differential equation in $\mathbb{R}$ given by $\ddot{z} + \sin(z) + \dot{z} = 0$, or in first-order form in $\mathbb{R}^2$

$$\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} = \begin{bmatrix} y_2 \\ -\sin(y_1) - y_2 \end{bmatrix} = f(\mathbf{y}).$$

The fixed points of the system are $\mathbf{p}_n = [n\pi, 0]^T$ for $n \in \mathbb{Z}$. Computation of the linearization yields

$$D_{\mathbf{y}}f = \begin{bmatrix} 0 & 1 \\ -\cos(y_1) & -1 \end{bmatrix}.$$

Hence, for the fixed points $\mathbf{p}_n$ with n even the eigenvalues are $\mu_1, \mu_2 = -\frac{1}{2} \pm i\sqrt{\frac{3}{4}}$ with negative real part (or Lyapunov exponent), while at the fixed points $\mathbf{p}_n$ with n odd one obtains as eigenvalues $\nu_1, \nu_2 = -\frac{1}{2} \pm \sqrt{\frac{5}{4}}$, resulting in one positive and one negative eigenvalue. Hence, the flow of the differential equation is locally C^0 -conjugate around all fixed points with even n , and around all fixed points with odd n , while the flows around, e.g., $\mathbf{p}_0$ and $\mathbf{p}_1$ are not conjugate.

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57

Control Theory

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Given a dynamical system described by the ordinary differential equation (ODE)

$$\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t)), \quad \mathbf{x}(t_0) = \mathbf{x}^0,$$

where $\mathbf{x}$ is the state of the system and $\mathbf{u}$ serves as input, the major problem in control theory is to steer the state from $\mathbf{x}^0$ to some desired state; i.e., for a given initial value $\mathbf{x}(t_0) = \mathbf{x}^0$ and target $\mathbf{x}^1$, can we find a piecewise continuous or L_2 (i.e., square-integrable, Lebesgue measurable) control function $\hat{\mathbf{u}}$ such that there exists $t_1 \geq t_0$ with $\mathbf{x}(t_1; \hat{\mathbf{u}}) = \mathbf{x}^1$, where $\mathbf{x}(t; \hat{\mathbf{u}})$ is the solution trajectory of the ODE given above for $\mathbf{u} \equiv \hat{\mathbf{u}}$? Often, the target is $\mathbf{x}^1 = 0$, in particular if $\mathbf{x}$ describes the deviation from a nominal path. A weaker demand is to asymptotically stabilize the system, i.e., to find an admissible control function $\hat{\mathbf{u}}$ (i.e., a piecewise continuous or L_2 function $\hat{\mathbf{u}} : [t_0, t_1] \mapsto \mathcal{U}$) such that $\lim_{t \rightarrow \infty} \mathbf{x}(t; \hat{\mathbf{u}}) = 0$.

Another major problem in control theory arises from the fact that often, not all states are available for measurements or observations. Thus, we are faced with the question: Given partial information about the states, is it possible to reconstruct the solution trajectory from the measurements/observations? If this is the case, the states can be estimated by state observers. The classical approach leads to the Luenberger observer, but nowadays most frequently the famous Kalman–Bucy filter [KB61] is used as it can be considered as an optimal state observer in a least-squares sense and allows for stochastic uncertainties in the system.

Analyzing the above questions concerning controllability, observability, etc. for general control systems is beyond the scope of linear algebra. Therefore, we will mostly focus on linear time-invariant (LTI) systems that can be analyzed with tools relying on linear algebra techniques. (For further reading, see, e.g., [Lev96], [Mut99], and [Son98].)

Once the above questions are settled, it is interesting to ask how the desired control objectives can be achieved in an optimal way. The linear-quadratic regulator (LQR) problem is equivalent to a dynamic optimization problem for linear differential equations. Its significance for control theory was fully discovered first by Kalman in 1960 [Kal60]. One of its main applications is to steer the solution of the underlying linear differential equation to a desired reference trajectory with minimal cost given full information on the states. If full information is not available, then the states can be estimated from the measurements or observations using a Kalman–Bucy filter. This leads to the linear-quadratic Gaussian (LQG) control problem. The latter problem and its solution were first described in the classical papers [Kal60] and [KB61] and are nowadays contained in any textbook on control theory.

In the past decades, the interest has shifted from optimal control to robust control. The question raised is whether a given control law is still able to achieve a desired performance in the presence of uncertain disturbances. In this sense, the LQR control law has some robustness, while the LQG design cannot be considered to be robust [Doy78]. The H_∞ control problem aims at minimizing the worst-case error that can occur if the system is perturbed by exogenous perturbations. It is, thus, one example of a *robust control* problem. We will only introduce the standard H_∞ control problem, though there exist many other robust control problems and several variations of the H_∞ control problem; see [GL95], [PUA00], [ZDG96].

Many of the above questions lead to methods that involve the solution of linear and nonlinear matrix equations, in particular Lyapunov, Sylvester, and Riccati equations. For instance, stability, controllability, and observability of LTI systems can be related to solutions of Lyapunov equations (see, e.g., [LT85, Sec. 13] and [HJ91]), while the LQR, LQG, and H_∞ control problems lead to the solution of algebraic Riccati equations, (see, e.g., [AKFIJ03], [Dat04], [LR95], [Meh91], and [Sim96]). Therefore, we will provide the most relevant properties of these matrix equations.

The concepts and solution techniques contained in this chapter and many other control-related algorithms are implemented in the MATLAB[®] Control System Toolbox, the Subroutine Library in Control SLICOT [BMS⁺99], and many other computer-aided control systems design tools.

Finally, we note that all concepts described in this section are related to *continuous-time* systems. Analogous concepts hold for *discrete-time* systems whose dynamics are described by difference equations; see, e.g., [Kuc91].

57.1 Basic Concepts

Definitions:

Given vector spaces $\mathcal{X}$ (the **state space**), $\mathcal{U}$ (the **input space**), and $\mathcal{Y}$ (the **output space**) and measurable functions $\mathbf{f}, \mathbf{g} : [t_0, t_f] \times \mathcal{X} \times \mathcal{U} \mapsto \mathbb{R}^n$, a **control system** is defined by

$$\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t)),$$

$$\mathbf{y}(t) = \mathbf{g}(t, \mathbf{x}(t), \mathbf{u}(t)),$$

where the differential equation is called the **state equation**, the second equation is called the **observer equation**, and $t \in [t_0, t_f]$ ($t_f \in [0, \infty]$).

Here,

$\mathbf{x} : [t_0, t_f] \mapsto \mathcal{X}$ is the **state (vector)**,

$\mathbf{u} : [t_0, t_f] \mapsto \mathcal{U}$ is the **control (vector)**,

$\mathbf{y} : [t_0, t_f] \mapsto \mathcal{Y}$ is the **output (vector)**.

A control system is called **autonomous (time-invariant)** if

$$\mathbf{f}(t, \mathbf{x}, \mathbf{u}) \equiv \mathbf{f}(\mathbf{x}, \mathbf{u}) \text{ and } \mathbf{g}(t, \mathbf{x}, \mathbf{u}) \equiv \mathbf{g}(\mathbf{x}, \mathbf{u}).$$

The number of state-space variables n is called the **order** or **degree** of the system.

Let $\mathbf{x}^1 \in \mathbb{R}^n$. A control system with initial value $\mathbf{x}(t_0) = \mathbf{x}^0$ is **controllable to $\mathbf{x}^1$ in time $t_1 > t_0$** if there exists an **admissible** control function $\mathbf{u}$ (i.e., a piecewise continuous or L_2 function $\mathbf{u} : [t_0, t_1] \mapsto \mathcal{U}$) such that $\mathbf{x}(t_1; \mathbf{u}) = \mathbf{x}^1$. (Equivalently, $(t_1, \mathbf{x}^1)$ is **reachable from** $(t_0, \mathbf{x}^0)$.)

A control system with initial value $\mathbf{x}(t_0) = \mathbf{x}^0$ is **controllable to $\mathbf{x}^1$** if there exists $t_1 > t_0$ such that $(t_1, \mathbf{x}^1)$ is reachable from $(t_0, \mathbf{x}^0)$.

If the control system is controllable to all $\mathbf{x}^1 \in \mathcal{X}$ for all $(t_0, \mathbf{x}^0)$ with $\mathbf{x}^0 \in \mathcal{X}$, it is **(completely) controllable**.

A control system is **linear** if $\mathcal{X} = \mathbb{R}^n$, $\mathcal{U} = \mathbb{R}^m$, $\mathcal{Y} = \mathbb{R}^p$, and

$$\mathbf{f}(t, \mathbf{x}, \mathbf{u}) = A(t)\mathbf{x}(t) + B(t)\mathbf{u}(t),$$

$$\mathbf{g}(t, \mathbf{x}, \mathbf{u}) = C(t)\mathbf{x}(t) + D(t)\mathbf{u}(t),$$

where $A : [t_0, t_f] \mapsto \mathbb{R}^{n \times n}$, $B : [t_0, t_f] \mapsto \mathbb{R}^{n \times m}$, $C : [t_0, t_f] \mapsto \mathbb{R}^{p \times n}$, $D : [t_0, t_f] \mapsto \mathbb{R}^{p \times m}$ are smooth functions.

A linear **time-invariant system (LTI system)** has the form

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + B\mathbf{u}(t),$$

$$\mathbf{y}(t) = C\mathbf{x}(t) + D\mathbf{u}(t),$$

with $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$, and $D \in \mathbb{R}^{p \times m}$.

An LTI system is **(asymptotically) stable** if the corresponding linear homogeneous ODE $\dot{\mathbf{x}} = A\mathbf{x}$ is (asymptotically) stable. (For a definition of (asymptotic) stability confer Chapter 55 and Chapter 56.)

An LTI system is **stabilizable (by state feedback)** if there exists an admissible control in the form of a **state feedback**

$$\mathbf{u}(t) = F\mathbf{x}(t), \quad F \in \mathbb{R}^{m \times n},$$

such that the unique solution of the corresponding **closed-loop** ODE

$$\dot{\mathbf{x}}(t) = (A + BF)\mathbf{x}(t) \tag{57.1}$$

is asymptotically stable.

An LTI system is **observable (reconstructible)** if for two solution trajectories $\mathbf{x}(t)$ and $\tilde{\mathbf{x}}(t)$ of its state equation, it holds that

$$C\mathbf{x}(t) = C\tilde{\mathbf{x}}(t) \quad \forall t \leq t_0 \quad (\forall t \geq t_0)$$

implies $\mathbf{x}(t) = \tilde{\mathbf{x}}(t) \quad \forall t \leq t_0 \quad (\forall t \geq t_0)$.

An LTI system is **detectable** if for any solution $\mathbf{x}(t)$ of $\dot{\mathbf{x}} = A\mathbf{x}$ with $C\mathbf{x}(t) \equiv 0$ we have $\lim_{t \rightarrow \infty} \mathbf{x}(t) = 0$.

Facts:

1. For LTI systems, all controllability and reachability concepts are equivalent. Therefore, we only speak of controllability of LTI systems.
2. Observability implies that one can obtain all necessary information about the LTI system from the output equation.
3. Detectability weakens observability in the same sense as stabilizability weakens controllability: Not all of $\mathbf{x}$ can be observed, but the unobservable part is asymptotically stable.
4. Observability (detectability) and controllability (stabilizability) are dual concepts in the following sense: an LTI system is observable (detectable) if and only if the dual system

$$\dot{\mathbf{z}}(t) = A^T \mathbf{z}(t) + C^T \mathbf{v}(t)$$

is controllable (stabilizable). This fact is sometimes called the **duality principle of control theory**.

Examples:

1. A fundamental problem in robotics is to control the position of a single-link rotational joint using a motor placed at the “pivot.” A simple mathematical model for this is the pendulum [Son98]. Applying a torque $\mathbf{u}$ as external force, this can serve as a means to control the motion of the pendulum (Figure 57.1).

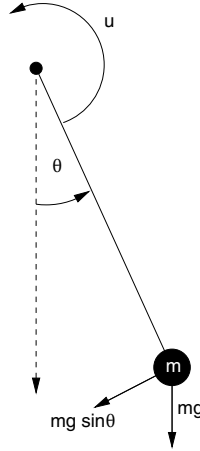


FIGURE 57.1 Pendulum as mathematical model of a single-link rotational joint .

If we neglect friction and assume that the mass is concentrated at the tip of the pendulum, Newton's law for rotating objects

$$m\ddot{\Theta}(t) + mg \sin \Theta(t) = \mathbf{u}(t)$$

describes the counter clockwise movement of the angle between the vertical axis and the pendulum subject to the control $\mathbf{u}(t)$. This is a first example of a (nonlinear) control system if we set

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \Theta(t) \\ \dot{\Theta}(t) \end{bmatrix},$$

$$\mathbf{f}(t, \mathbf{x}, \mathbf{u}) = \begin{bmatrix} x_2 \\ -mg \sin(x_1) \end{bmatrix}, \quad \mathbf{g}(t, \mathbf{x}, \mathbf{u}) = x_1,$$

where we assume that only $\Theta(t)$ can be measured, but not the angular velocity $\dot{\Theta}(t)$.

For $\mathbf{u}(t) \equiv 0$, the stationary position $\Theta = \pi, \dot{\Theta} = 0$ is an unstable equilibrium, i.e., small perturbations will lead to unstable motion. The objective now is to apply a torque (control $\mathbf{u}$) to correct for deviations from this unstable equilibrium, i.e., to keep the pendulum in the upright position, (Figure 57.2).

2. Scaling the variables such that $m = 1 = g$ and assuming a small perturbation $\Theta - \pi$ in the inverted pendulum problem described above, we have

$$\sin \Theta = -(\Theta - \pi) + o((\Theta - \pi)^2).$$

(Here, $\mathbf{g}(x) = o(x)$ if $\lim_{x \rightarrow \infty} \frac{\mathbf{g}(x)}{x} = 0$.) This allows us to linearize the control system in order to obtain a linear control system for $\varphi(t) := \Theta(t) - \pi$:

$$\ddot{\varphi}(t) - \varphi(t) = \mathbf{u}(t).$$

This can be written as an LTI system, assuming only positions can be observed, with

$$\mathbf{x} = \begin{bmatrix} \varphi \\ \dot{\varphi} \end{bmatrix}, \quad A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

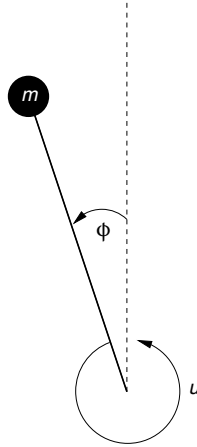


FIGURE 57.2 Inverted pendulum; apply control to move to upright position.

Now the objective translates to: Given initial values $x_1(0) = \varphi(0)$, $x_2(0) = \dot{\varphi}(0)$, find $\mathbf{u}(t)$ to bring $\mathbf{x}(t)$ to zero “as fast as possible.” It is usually an additional goal to avoid overshoot and oscillating behavior as much as possible.

57.2 Frequency-Domain Analysis

So far, LTI systems are treated in state-space. In systems and control theory, it is often beneficial to use the **frequency domain** formalism obtained from applying the Laplace transformation to its state and observer equations.

Definitions:

The rational matrix function

$$G(s) = C(sI - A)^{-1}B + D \in \mathbb{R}^{p \times m}[s]$$

is called the **transfer function** of the LTI system defined in section 57.1.

In a **frequency domain analysis**, $G(s)$ is evaluated for $s = i\omega$, where $\omega \in [0, \infty]$ has the physical interpretation of a frequency and the input is considered as a signal with frequency ω .

The L_∞ -**norm** of a transfer function is the operator norm induced by the frequency domain analogue of the L_2 -norm that applies to Laplace transformed input functions $\mathbf{u} \in L_2(-\infty, \infty; \mathbb{R}^m)$, where $L_2(a, b; \mathbb{R}^m)$ is the Lebesgue space of square-integrable, measurable functions on the interval $(a, b) \subset \mathbb{R}$ with values in $\mathbb{R}^m$.

The $p \times m$ -matrix-valued functions G for which $\|G\|_{L_\infty}$ is bounded form the space L_∞ .

The subset of L_∞ containing all $p \times m$ -matrix-valued functions that are analytical and bounded in the open right-half complex plane form the **Hardy space** H_∞ .

The H_∞ -**norm** of $G \in H_\infty$ is defined as

$$\|G\|_{H_\infty} = \operatorname{ess\,sup}_{\omega \in \mathbb{R}} \sigma_{\max}(G(i\omega)), \quad (57.2)$$

where $\sigma_{\max}(M)$ is the maximum singular value of the matrix M and $\operatorname{ess\,sup}_{t \in M} h(t)$ is the essential supremum of a function h evaluated on the set M , which is the function’s supremum on $M \setminus L$ where L is a set of Lebesgue measure zero.

For $T \in \mathbb{R}^{n \times n}$ nonsingular, the mapping implied by

$$(A, B, C, D) \mapsto (TAT^{-1}, TB, CT^{-1}, D)$$

is called a **state-space transformation**.

(A, B, C, D) is called a **realization** of an LTI system if its transfer function can be expressed as $G(s) = C(sI_n - A)^{-1}B + D$.

The minimum number $\hat{n}$ so that there exists no realization of a given LTI system with $n < \hat{n}$ is called the **McMillan degree** of the system.

A realization with $n = \hat{n}$ is a **minimal realization**.

Facts:

1. If $\mathbf{X}, \mathbf{Y}, \mathbf{U}$ are the Laplace transforms of $\mathbf{x}, \mathbf{y}, \mathbf{u}$, respectively, s is the Laplace variable, and $\mathbf{x}(0) = 0$, the state and observer equation of an LTI system transform to

$$\begin{aligned} s\mathbf{X}(s) &= A\mathbf{X}(s) + B\mathbf{U}(s), \\ \mathbf{Y}(s) &= C\mathbf{X}(s) + D\mathbf{U}(s). \end{aligned}$$

Thus, the resulting input–output relation

$$\mathbf{Y}(s) = (C(sI - A)^{-1}B + D)\mathbf{U}(s) = G(s)\mathbf{U}(s) \quad (57.3)$$

is completely determined by the transfer function of the LTI system.

2. As a consequence of the maximum modulus theorem, H_∞ functions must be bounded on the imaginary axis so that the essential supremum in the definition of the H_∞ -norm simplifies to a supremum for rational functions G .
3. The transfer function of an LTI system is invariant w.r.t. state-space transformations:

$$D + (CT^{-1})(sI - TAT^{-1})^{-1}(TB) = C(sI_n - A)^{-1}B + D = G(s).$$

Consequently, there exist infinitely many realizations of an LTI system.

4. Adding zero inputs/outputs does not change the transfer function, thus the order n of the system can be increased arbitrarily.

Examples:

1. The LTI system corresponding to the inverted pendulum has the transfer function

$$G(s) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s & -1 \\ -1 & s \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + [0] = \frac{1}{s^2 - 1}.$$

2. The L_∞ -norm of the transfer function corresponding to the inverted pendulum is

$$\|G\|_{L_\infty} = 1.$$

3. The transfer function corresponding to the inverted pendulum is not in H_∞ as $G(s)$ has a pole at $s = 1$ and, thus, is not bounded in the right-half plane.

57.3 Analysis of LTI Systems

In this section, we provide characterizations of the properties of LTI systems defined in the introduction. Controllability and the related concepts can be checked using several algebraic criteria.

Definitions:

A matrix $A \in \mathbb{R}^{n \times n}$ is **Hurwitz** or **(asymptotically) stable** if all its eigenvalues have strictly negative real part.

The **controllability matrix** corresponding to an LTI system is

$$\mathcal{C}(A, B) = [B, AB, A^2B, \dots, A^{n-1}B] \in \mathbb{R}^{n \times n \cdot m}.$$

The **observability matrix** corresponding to an LTI system is

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \in \mathbb{R}^{np \times n}.$$

The following transformations are **state-space transformations**:

- **Change of basis:**

$$\begin{aligned} \mathbf{x} &\mapsto P\mathbf{x} & \text{for } P \in \mathbb{R}^{n \times n} & \text{ nonsingular,} \\ \mathbf{u} &\mapsto Q\mathbf{u} & \text{for } Q \in \mathbb{R}^{m \times m} & \text{ nonsingular,} \\ \mathbf{y} &\mapsto R\mathbf{y} & \text{for } R \in \mathbb{R}^{p \times p} & \text{ nonsingular.} \end{aligned}$$

- **Linear state feedback:** $\mathbf{u} \mapsto F\mathbf{x} + \mathbf{v}$, $F \in \mathbb{R}^{m \times n}$, $\mathbf{v} : [t_0, t_f] \mapsto \mathbb{R}^m$.
- **Linear output feedback:** $\mathbf{u} \mapsto G\mathbf{y} + \mathbf{v}$, $G \in \mathbb{R}^{m \times p}$, $\mathbf{v} : [t_0, t_f] \mapsto \mathbb{R}^m$.

The **Kalman decomposition** of (A, B) is

$$V^T A V = \begin{bmatrix} A_1 & A_2 \\ 0 & A_3 \end{bmatrix}, \quad V^T B = \begin{bmatrix} B_1 \\ 0 \end{bmatrix}, \quad V \in \mathbb{R}^{n \times n} \text{ orthogonal,}$$

where (A_1, B_1) is controllable.

The **observability Kalman decomposition** of (A, C) is

$$W^T A W = \begin{bmatrix} A_1 & 0 \\ A_2 & A_3 \end{bmatrix}, \quad C W = [C_1 \ 0], \quad W \in \mathbb{R}^{n \times n} \text{ orthogonal,}$$

where (A_1, C_1) is observable.

Facts:

1. An LTI system is asymptotically stable if and only if A is Hurwitz.
2. For a given LTI system, the following are equivalent.
 - (a) The LTI system is controllable.
 - (b) The controllability matrix corresponding to the LTI system has full (row) rank, i.e., $\text{rank } \mathcal{C}(A, B) = n$.

(c) (**Hautus–Popov test**) If $\mathbf{p} \neq 0$ and $\mathbf{p}^* A = \lambda \mathbf{p}^*$, then $\mathbf{p}^* B \neq 0$.

(d) $\text{rank}([\lambda I - A, B]) = n \quad \forall \lambda \in \mathbb{C}$.

The essential part of the proof of the above characterizations (which is “d) $\Rightarrow$ b)”) is an application of the Cayley–Hamilton theorem (Section 4.3).

3. For a given LTI system, the following are equivalent:

(a) The LTI system is stabilizable, i.e., $\exists F \in \mathbb{R}^{m \times n}$ such that $A + BF$ is Hurwitz.

(b) (**Hautus–Popov test**) If $\mathbf{p} \neq 0$, $\mathbf{p}^* A = \lambda \mathbf{p}^*$, and $\text{Re}(\lambda) \geq 0$, then $\mathbf{p}^* B \neq 0$.

(c) $\text{rank}([A - \lambda I, B]) = n, \quad \forall \lambda \in \mathbb{C} \text{ with } \text{Re}(\lambda) \geq 0$.

(d) In the Kalman decomposition of (A, B) , A_3 is Hurwitz.

4. Using the change of basis $\tilde{\mathbf{x}} = V^T \mathbf{x}$ implied by the Kalman decomposition, we obtain

$$\begin{aligned}\dot{\tilde{\mathbf{x}}}_1 &= A_1 \tilde{\mathbf{x}}_1 + A_2 \tilde{\mathbf{x}}_2 + B_1 \mathbf{u}, \\ \dot{\tilde{\mathbf{x}}}_2 &= A_3 \tilde{\mathbf{x}}_2.\end{aligned}$$

Thus, $\tilde{\mathbf{x}}_2$ is not controllable. The eigenvalues of A_3 are, therefore, called **uncontrollable modes**.

5. For a given LTI system, the following are equivalent:

(a) The LTI system is observable.

(b) The observability matrix corresponding to the LTI system has full (column) rank, i.e., $\text{rank } \mathcal{O}(A, C) = n$.

(c) (**Hautus–Popov test**), $\mathbf{p} \neq 0$, $A\mathbf{p} = \lambda \mathbf{p} \implies C^T \mathbf{p} \neq 0$.

(d) $\text{rank} \begin{bmatrix} \lambda I - A \\ C \end{bmatrix} = n, \quad \forall \lambda \in \mathbb{C}$.

6. For a given LTI system, the following are equivalent:

(a) The LTI system is detectable.

(b) The dual system $\dot{\mathbf{z}} = A^T \mathbf{z} + C^T \mathbf{v}$ is stabilizable.

(c) (**Hautus–Popov test**) $\mathbf{p} \neq 0$, $A\mathbf{p} = \lambda \mathbf{p}$, $\text{Re}(\lambda) \geq 0 \implies C^T \mathbf{p} \neq 0$.

(d) $\text{rank} \begin{bmatrix} \lambda I - A \\ C \end{bmatrix} = n, \quad \forall \lambda \in \mathbb{C} \text{ with } \text{Re}(\lambda) \geq 0$.

(e) In the observability Kalman decomposition of (A, C) , A_3 is Hurwitz.

7. Using the change of basis $\tilde{\mathbf{x}} = W^T \mathbf{x}$ implied by the observability Kalman decomposition we obtain

$$\begin{aligned}\dot{\tilde{\mathbf{x}}}_1 &= A_1 \tilde{\mathbf{x}}_1 + B_1 \mathbf{u}, \\ \dot{\tilde{\mathbf{x}}}_2 &= A_2 \tilde{\mathbf{x}}_1 + A_3 \tilde{\mathbf{x}}_2 + B_2 \mathbf{u}, \\ \mathbf{y} &= C_1 \tilde{\mathbf{x}}_1.\end{aligned}$$

Thus, $\tilde{\mathbf{x}}_2$ is not observable. The eigenvalues of A_3 are, therefore, called **unobservable modes**.

8. The characterizations of observability and detectability are proved using the duality principle and the characterizations of controllability and stabilizability.
9. If an LTI system is controllable (observable, stabilizable, detectable), then the corresponding LTI system resulting from a state-space transformation is controllable (observable, stabilizable, detectable).

10. For $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$ there exist $P \in \mathbb{R}^{n \times n}$, $Q \in \mathbb{R}^{m \times m}$ orthogonal such that

$$PAP^T = \begin{bmatrix} A_{11} & & & A_{1,s-1} & & A_{1,s} \\ A_{21} & \ddots & & \vdots & & \vdots \\ 0 & \ddots & \ddots & \vdots & & \vdots \\ \vdots & \ddots & & & & \\ 0 & \cdots & 0 & A_{s-1,s-2} & A_{s-1,s-1} & A_{s-1,s} \\ 0 & \cdots & 0 & 0 & 0 & A_{s,s} \end{bmatrix} \begin{matrix} n_1 \\ n_2 \\ \\ \\ n_{s-1} \\ n_s \end{matrix},$$

$$PBQ = \begin{bmatrix} B_1 & 0 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \end{bmatrix} \begin{matrix} n_1 \\ n_2 \\ \vdots \\ n_s \end{matrix},$$

$$n_1 \quad n_{s-2} \quad n_{s-1} \quad n_s$$

where $n_1 \geq n_2 \geq \dots \geq n_{s-1} \geq n_s \geq 0$, $n_{s-1} > 0$, $A_{i,i-1} = [\Sigma_{i,i-1} \ 0] \in \mathbb{R}^{n_i \times n_{i-1}}$, $\Sigma_{i,i-1} \in \mathbb{R}^{n_i \times n_i}$ nonsingular for $i = 1, \dots, s-1$, $\Sigma_{s-1,s-2}$ is diagonal, and B_1 is nonsingular.

Moreover, this transformation to **staircase form** can be computed by a finite sequence of singular value decompositions.

11. An LTI system is controllable if in the staircase form of (A, B) , $n_s = 0$.
12. An LTI system is observable if $n_s = 0$ in the staircase form of (A^T, C^T) .
13. An LTI system is stabilizable if in the staircase form of (A, B) , $A_{s,s}$ is Hurwitz.
14. An LTI system is detectable if in the staircase form of (A^T, C^T) , $A_{s,s}$ is Hurwitz.
15. In case $m = 1$, the staircase form of (A, B) is given by

$$PAP^T = \begin{bmatrix} a_{11} & \cdots & \cdots & a_{1,n} \\ a_{21} & & & \vdots \\ & \ddots & & \vdots \\ & & a_{n,n-1} & a_{n,n} \end{bmatrix}, PB = \begin{bmatrix} b_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

and is called the **controllability Hessenberg form**. The corresponding staircase form of (A^T, C^T) in case $p = 1$ is called the **observability Hessenberg form**.

Examples:

1. The LTI system corresponding to the inverted pendulum problem is not asymptotically stable as A is not Hurwitz: $\sigma(A) = \{\pm 1\}$.
2. The LTI system corresponding to the inverted pendulum problem is controllable as the controllability matrix

$$\mathcal{C}(A, B) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

has full rank. Thus, it is also stabilizable.

3. The LTI system corresponding to the inverted pendulum problem is observable as the observability matrix

$$\mathcal{O}(A, C) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

has full rank. Thus, it is also detectable.

57.4 Matrix Equations

A fundamental role in many tasks in control theory is played by matrix equations. We, therefore, review their most important properties. More details can be found in [AKFIJ03], [HJ91], [LR95], and [LT85].

Definitions:

A linear matrix equation of the form

$$AX + XB = W, \quad A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{m \times m}, W \in \mathbb{R}^{n \times m},$$

is called **Sylvester equation**.

A linear matrix equation of the form

$$AX + XA^T = W, \quad A \in \mathbb{R}^{n \times n}, W = W^T \in \mathbb{R}^{n \times n},$$

is called **Lyapunov equation**.

A quadratic matrix equation of the form

$$0 = Q + A^T X + XA - XGX, \quad A \in \mathbb{R}^{n \times n}, G = G^T, Q = Q^T \in \mathbb{R}^{n \times n},$$

is called **algebraic Riccati equation (ARE)**.

Facts:

1. The Sylvester equation is equivalent to the linear system of equations

$$[(I_m \otimes A) + (B^T \otimes I_n)] \text{vec}(X) = \text{vec}(W),$$

where $\otimes$ and vec denote the Kronecker product and the vec -operator defined in Section 10.4. Thus, the Sylvester equation has a unique solution if and only if $\sigma(A) \cap \sigma(-B) = \emptyset$.

2. The Lyapunov equation is equivalent to the linear system of equations

$$[(I_m \otimes A) + (A \otimes I_n)] \text{vec}(X) = \text{vec}(W).$$

Thus, it has a unique solution if and only if $\sigma(A) \cap \sigma(-A^T) = \emptyset$. In particular, this holds if A is Hurwitz.

3. If G and Q are positive semidefinite with (A, G) stabilizable and (A, Q) detectable, then the ARE has a unique positive semidefinite solution X_* with the property that $\sigma(A - GX_*)$ is Hurwitz.
4. If the assumptions given above are not satisfied, there may or may not exist a stabilizing solution with the given properties. Besides, there may exist a continuum of solutions, a finite number of solutions, or no solution at all. The solution theory for AREs is a vast topic by itself; see the monographs [AKFIJ03], [LR95] and [Ben99], [Dat04], [Meh91], and [Sim96] for numerical algorithms to solve these equations.

Examples:

1. For

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}, \quad W = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix},$$

a solution of the Sylvester equation is

$$X = \frac{1}{4} \begin{bmatrix} -3 & 3 \\ 1 & -3 \end{bmatrix}.$$

Note that $\sigma(A) = \sigma(B) = \{1, 1\}$ so that $\sigma(A) \cap \sigma(-B) = \emptyset$. Thus, this Sylvester equation has the unique solution X given above.

2. For

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix},$$

the stabilizing solution of the associated ARE is

$$X_* = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

and the spectrum of the closed-loop matrix

$$A - GX_* = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$$

is $\{-1, -1\}$.

3. Consider the ARE

$$0 = C^T C + A^T X + X A - X B B^T X$$

corresponding to an LTI system with

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C = [\sqrt{2} \quad 0], \quad D = 0.$$

For this ARE, $X = \begin{bmatrix} -1 + \sqrt{3} & 0 \\ 0 & \xi \end{bmatrix}$ is a solution for all $\xi \in \mathbb{R}$. It is positive semidefinite for all $\xi \geq 0$, but this ARE does not have a stabilizing solution as the LTI system is neither stabilizable nor detectable.

57.5 State Estimation

In this section, we present the two most famous approaches to state observation, that is, finding a function $\hat{\mathbf{x}}(t)$ that approximates the state $\mathbf{x}(t)$ of a given LTI system if only its inputs $\mathbf{u}(t)$ and outputs $\mathbf{y}(t)$ are known. While the first approach (the Luenberger observer) assumes a deterministic system behavior, the Kalman–Bucy filter allows for uncertainty in the system, modeled by white-noise, zero-mean stochastic processes.

Definitions:

Given an LTI system with $D = 0$, a **state observer** is a function

$$\hat{\mathbf{x}} : [0, \infty) \mapsto \mathbb{R}^n$$

such that for some nonsingular matrix $Z \in \mathbb{R}^{n \times n}$ and $\mathbf{e}(t) = \hat{\mathbf{x}}(t) - Z\mathbf{x}(t)$, we have

$$\lim_{t \rightarrow \infty} \mathbf{e}(t) = 0.$$

Given an LTI system with stochastic disturbances

$$\begin{aligned}\dot{\mathbf{x}}(t) &= A\mathbf{x}(t) + B\mathbf{u}(t) + \tilde{B}\mathbf{w}(t), \\ \mathbf{y}(t) &= C\mathbf{x}(t) + \mathbf{v}(t),\end{aligned}$$

where A, B, C are as before, $\tilde{B} \in \mathbb{R}^{n \times \tilde{m}}$, and $\mathbf{w}(t), \mathbf{v}(t)$ are white-noise, zero-mean stochastic processes with corresponding covariance matrices $W = W^T \in \mathbb{R}^{\tilde{m} \times \tilde{m}}$ (positive semidefinite), $V = V^T \in \mathbb{R}^{p \times p}$ (positive definite), the problem to minimize the mean square error

$$E [\|\mathbf{x}(t) - \hat{\mathbf{x}}(t)\|_2^2]$$

over all state observers is called the **optimal estimation problem**. (Here, $E[r]$ is the **expected value** of r .)

Facts:

1. A state observer, called the **Luenberger observer**, is obtained as the solution of the dynamical system

$$\dot{\hat{\mathbf{x}}}(t) = H\hat{\mathbf{x}}(t) + F\mathbf{y}(t) + G\mathbf{u}(t),$$

where $H \in \mathbb{R}^{n \times n}$ and $F \in \mathbb{R}^{n \times p}$ are chosen so that H is Hurwitz and the **Sylvester observer equation**

$$HX - XA + FC = 0$$

has a nonsingular solution X . Then $G = XB$ and the matrix Z in the definition of the state observer equals the solution X of the Sylvester observer equation.

2. Assuming that

- $\mathbf{w}$ and $\mathbf{v}$ are uncorrelated stochastic processes,
- the initial state $\mathbf{x}^0$ is a Gaussian zero-mean random variable, uncorrelated with $\mathbf{w}$ and $\mathbf{v}$,
- (A, B) is controllable and (A, C) is observable,

the solution to the optimal estimation problem is given by the **Kalman–Bucy filter**, defined as the solution of the linear differential equation

$$\dot{\hat{\mathbf{x}}}(t) = (A - Y_*C^TV^{-1}C)\hat{\mathbf{x}}(t) + B\mathbf{u}(t) + Y_*C^TV^{-1}\mathbf{y}(t),$$

where Y_* is the unique stabilizing solution of the **filter ARE**:

$$0 = \tilde{B}W\tilde{B}^T + AY + YA^T - YC^TV^{-1}CY.$$

3. Under the same assumptions as above, the stabilizing solution of the filter ARE can be shown to be symmetric positive definite.

Examples:

1. A Luenberger observer for the LTI system corresponding to the inverted pendulum problem can be constructed as follows: Choose $H = \text{diag}(-2, -\frac{1}{2})$ and $F = \begin{bmatrix} 2 & 1 \end{bmatrix}^T$. Then the Sylvester observer equation has the unique solution

$$X = \frac{1}{3} \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}.$$

Note that X is nonsingular. Thus, we get $G = XB = \frac{1}{3} \begin{bmatrix} -2 & 4 \end{bmatrix}$.

2. Consider the inverted pendulum with disturbances v, w and $\tilde{B} = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$. Assume that $V = W = 1$. The Kalman–Bucy filter is determined via the filter ARE, yielding

$$Y_* = (1 + \sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Thus, the state estimation obtained from the Kalman filter is given by the solution of

$$\dot{\hat{\mathbf{x}}}(t) = \begin{bmatrix} -1 - \sqrt{2} & 1 \\ -\sqrt{2} & 0 \end{bmatrix} \hat{\mathbf{x}}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mathbf{u}(t) + (1 + \sqrt{2}) \begin{bmatrix} 1 \\ 1 \end{bmatrix} \mathbf{y}(t).$$

57.6 Control Design for LTI Systems

This section provides the background for some of the most important control design methods.

Definitions:

A **(feedback) controller** for an LTI system is given by another LTI system

$$\begin{aligned} \dot{\mathbf{r}}(t) &= E\mathbf{r}(t) + F\mathbf{y}(t), \\ \mathbf{u}(t) &= H\mathbf{r}(t) + K\mathbf{y}(t), \end{aligned}$$

where $E \in \mathbb{R}^{N \times N}$, $F \in \mathbb{R}^{N \times p}$, $H \in \mathbb{R}^{m \times N}$, $K \in \mathbb{R}^{m \times p}$, and the “output” $\mathbf{u}(t)$ of the controller serves as the input for the original LTI system.

If E, F, H are zero matrices, a controller is called **static feedback**, otherwise it is called a **dynamic compensator**.

A static feedback control law is a **state feedback** if in the controller equations, the output function $\mathbf{y}(t)$ is replaced by the state $\mathbf{x}(t)$, otherwise it is called **output feedback**.

The **closed-loop system** resulting from inserting the control law $\mathbf{u}(t)$ obtained from a dynamic compensator into the LTI system is illustrated by the block diagram in Figure 57.3, where $\mathbf{w}$ is as in the definition of LTI systems with stochastic disturbances in Section 57.5 and $\mathbf{z}$ will only be needed later when defining the H_∞ control problem.

The linear-quadratic optimization (optimal control) problem

$$\min_{\mathbf{u} \in L_2(0, \infty; \mathcal{U})} \mathcal{J}(\mathbf{u}), \quad \text{where } \mathcal{J}(\mathbf{u}) = \frac{1}{2} \int_0^\infty (\mathbf{y}(t)^T Q \mathbf{y}(t) + \mathbf{u}(t)^T R \mathbf{u}(t)) dt$$

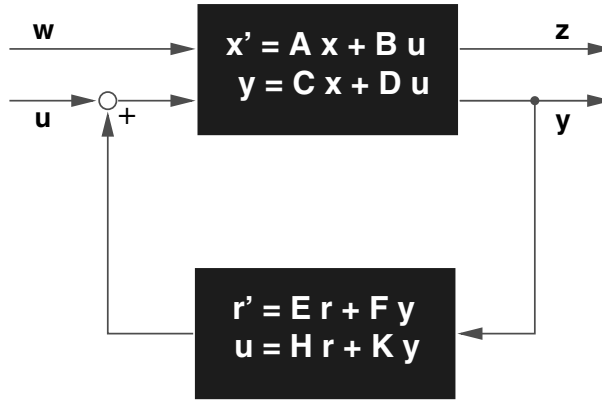


FIGURE 57.3 Closed-loop diagram of an LTI system and a dynamic compensator.

subject to the dynamical constraint given by an LTI system is called the **linear-quadratic regulator (LQR) problem**.

The linear-quadratic optimization (optimal control) problem

$$\min_{\mathbf{u} \in L_2(0, \infty; \mathcal{U})} \mathcal{J}(\mathbf{u}), \quad \text{where } \mathcal{J}(\mathbf{u}) = \lim_{t_f \rightarrow \infty} \frac{1}{2t_f} E \left[\int_{-t_f}^{t_f} (\mathbf{y}(t)^T \mathbf{Q} \mathbf{y}(t) + \mathbf{u}(t)^T \mathbf{R} \mathbf{u}(t)) dt \right]$$

subject to the dynamical constraint given by an LTI system with stochastic disturbances is called the **linear-quadratic Gaussian (LQG) problem**.

Consider an LTI system where inputs and outputs are split into two parts, so that instead of $\mathbf{B}\mathbf{u}(t)$ we have

$$\mathbf{B}_1 \mathbf{w}(t) + \mathbf{B}_2 \mathbf{u}(t),$$

and instead of $\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$, we write

$$\mathbf{z}(t) = \mathbf{C}_1 \mathbf{x}(t) + \mathbf{D}_{11} \mathbf{w}(t) + \mathbf{D}_{12} \mathbf{u}(t),$$

$$\mathbf{y}(t) = \mathbf{C}_2 \mathbf{x}(t) + \mathbf{D}_{21} \mathbf{w}(t) + \mathbf{D}_{22} \mathbf{u}(t),$$

where $\mathbf{u}(t) \in \mathbb{R}^{m_2}$ denotes the control input, $\mathbf{w}(t) \in \mathbb{R}^{m_1}$ is an **exogenous input** that may include noise, linearization errors, and unmodeled dynamics, $\mathbf{y}(t) \in \mathbb{R}^{p_2}$ contains **measured outputs**, while $\mathbf{z}(t) \in \mathbb{R}^{p_1}$ is the **regulated output** or an **estimation error**. Let $\mathbf{G} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ denote the corresponding transfer function such that

$$\begin{bmatrix} \mathbf{Z} \\ \mathbf{Y} \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} \mathbf{W} \\ \mathbf{U} \end{bmatrix},$$

where $\mathbf{Y}, \mathbf{Z}, \mathbf{U}, \mathbf{W}$ denote the Laplace transforms of $\mathbf{y}, \mathbf{z}, \mathbf{u}, \mathbf{w}$.

The **optimal H_∞ control problem** is then to determine a dynamic compensator

$$\dot{\mathbf{r}}(t) = \mathbf{E} \mathbf{r}(t) + \mathbf{F} \mathbf{y}(t),$$

$$\mathbf{u}(t) = \mathbf{H} \mathbf{r}(t) + \mathbf{K} \mathbf{y}(t),$$

with $E \in \mathbb{R}^{N \times N}$, $F \in \mathbb{R}^{N \times p_2}$, $H \in \mathbb{R}^{m_2 \times N}$, $K \in \mathbb{R}^{m_2 \times p_2}$ and transfer function $M(s) = H(sI - E)^{-1}F + K$ such that the resulting closed-loop system

$$\begin{aligned}\dot{\mathbf{x}}(t) &= (A + B_2 K Z_1 C_2)\mathbf{x}(t) + (B_2 Z_2 H)\mathbf{r}(t) + (B_1 + B_2 K Z_1 D_{21})\mathbf{w}(t), \\ \dot{\mathbf{r}}(t) &= F Z_1 C_2 \mathbf{x}(t) + (E + F Z_1 D_{22} H)\mathbf{r}(t) + F Z_1 D_{21} \mathbf{w}(t), \\ \mathbf{z}(t) &= (C_1 + D_{12} Z_2 K C_2)\mathbf{x}(t) + D_{12} Z_2 H \mathbf{r}(t) + (D_{11} + D_{12} K Z_1 D_{21})\mathbf{w}(t),\end{aligned}$$

with $Z_1 = (I - D_{22}K)^{-1}$ and $Z_2 = (I - K D_{22})^{-1}$,

- is **internally stable**, i.e., the solution of the system with $\mathbf{w}(t) \equiv 0$ is asymptotically stable, and
- the closed-loop transfer function $T_{zw}(s) = G_{22}(s) + G_{21}(s)M(s)(I - G_{11}(s)M(s))^{-1}G_{12}(s)$ from $\mathbf{w}$ to $\mathbf{z}$ is minimized in the H_∞ -norm.

The **suboptimal H_∞ control problem** is to find an internally stabilizing controller so that

$$\|T_{zw}\|_{H_\infty} < \gamma,$$

where $\gamma > 0$ is a robustness threshold.

Facts:

1. If $D = 0$ and the LTI system is both stabilizable and detectable, the weighting matrix Q is positive semidefinite, and R is positive definite, then the solution of the LQR problem is given by the state feedback controller

$$\mathbf{u}_*(t) = -R^{-1}B^T X_* \mathbf{x}(t), \quad t \geq 0,$$

where X_* is the unique stabilizing solution of the LQR ARE,

$$0 = C^T Q C + A^T X + X A - X B R^{-1} B^T X.$$

2. The LQR problem does not require an observer equation — inserting $\mathbf{y}(t) = C\mathbf{x}(t)$ into the cost functional, we obtain a problem formulation depending only on states and inputs:

$$\begin{aligned}\mathcal{J}(\mathbf{u}) &= \frac{1}{2} \int_0^\infty (\mathbf{y}(t)^T Q \mathbf{y}(t) + \mathbf{u}(t)^T R \mathbf{u}(t)) \, dt \\ &= \frac{1}{2} \int_0^\infty (\mathbf{x}(t)^T C^T Q C \mathbf{x}(t) + \mathbf{u}(t)^T R \mathbf{u}(t)) \, dt.\end{aligned}$$

3. Under the given assumptions, it can also be shown that X_* is symmetric and the unique positive semidefinite matrix among all solutions of the LQR ARE.
4. The assumptions for the feedback solution of the LQR problem can be weakened in several aspects; see, e.g., [Gee89] and [SSC95].
5. Assuming that

- $\mathbf{w}$ and $\mathbf{v}$ are uncorrelated stochastic processes,
- the initial state $\mathbf{x}^0$ is a Gaussian zero-mean random variable, uncorrelated with $\mathbf{w}$ and $\mathbf{v}$,
- (A, B) is controllable and (A, C) is observable,

the solution to the LQG problem is given by the feedback controller

$$\mathbf{u}(t) = -R^{-1}B^T X_* \hat{\mathbf{x}}(t),$$

where X_* is the solution of the LQR ARE and $\hat{\mathbf{x}}$ is the Kalman–Bucy filter

$$\dot{\hat{\mathbf{x}}}(t) = (A - BR^{-1}B^T X_* - Y_* C^T V^{-1}C)\hat{\mathbf{x}}(t) + Y_* C^T V^{-1}\mathbf{y}(t),$$

corresponding to the closed-loop system resulting from the LQR solution with Y_* being the stabilizing solution of the corresponding filter ARE.

6. In principle, there is no restriction on the degree N of the H_∞ controller, although smaller dimensions N are preferred for practical implementation and computation.
7. The state-space solution to the H_∞ suboptimal control problem [DGKF89] relates H_∞ control to AREs: under the assumptions that
 - (A, B_k) is stabilizable and (A, C_k) is detectable for $k = 1, 2$,
 - $D_{11} = 0$, $D_{22} = 0$, and

$$D_{12}^T \begin{bmatrix} C_1 & D_{12} \end{bmatrix} = \begin{bmatrix} 0 & I \end{bmatrix}, \quad \begin{bmatrix} B_1 \\ D_{21} \end{bmatrix} D_{21}^T = \begin{bmatrix} 0 \\ I \end{bmatrix},$$

a suboptimal H_∞ controller exists if and only if the AREs

$$0 = C_1^T C_1 + AX + XA^T + X \left(\frac{1}{\gamma^2} B_1 B_1^T - B_2 B_2^T \right) X,$$

$$0 = B_1^T B_1 + A^T Y + YA + Y \left(\frac{1}{\gamma^2} C_1 C_1^T - C_2 C_2^T \right) Y$$

both have positive semidefinite stabilizing solutions X_∞ and Y_∞ , respectively, satisfying the spectral radius condition

$$\rho(XY) < \gamma^2.$$

8. The solution of the optimal H_∞ control problem can be obtained by a bisection method (or any other root-finding method) minimizing γ based on the characterization of an H_∞ suboptimal controller given in Fact 7, starting from γ_0 for which no suboptimal H_∞ controller exists and γ_1 for which the above conditions are satisfied.
9. The assumptions made for the state-space solution of the H_∞ control problem can mostly be relaxed.
10. The robust numerical solution of the H_∞ control problem is a topic of ongoing research — the solution via AREs may suffer from several difficulties in the presence of roundoff errors and should be avoided if possible. One way out is a reformulation of the problem using structured generalized eigenvalue problems; see [BBMX99b], [CS92] and [GL97].
11. Once a (sub-)optimal γ is found, it remains to determine a realization of the H_∞ controller. One possibility is the **central (minimum entropy) controller** [ZDG96]:

$$E = A + \frac{1}{\gamma^2} B_1 B_1^T - B_2 B_2^T X_\infty - Z_\infty Y_\infty C_2^T C_2,$$

$$F = Z_\infty Y_\infty C_2^T, \quad K = -B_2^T X_\infty, \quad H = 0,$$

where

$$Z_\infty = \left(I - \frac{1}{\gamma^2} Y_\infty X_\infty \right)^{-1}.$$

Examples:

1. The cost functional in the LQR and LQG problems values the energy needed to reach the desired state by the weighting matrix R on the inputs. Thus, usually

$$R = \text{diag}(\rho_1, \dots, \rho_m).$$

The weighting on the states or outputs in the LQR or LQG problems is usually used to penalize deviations from the desired state of the system and is often also given in diagonal form. Common examples of weighting matrices are $R = \rho I_m$, $Q = \gamma I_p$ for $\rho, \gamma > 0$.

2. The solution to the LQR problem for the inverted pendulum with $Q = R = 1$ is given via the stabilizing solution of the LQR ARE, which is

$$X_* = \begin{bmatrix} 2\sqrt{1+\sqrt{2}} & 1+\sqrt{2} \\ 1+\sqrt{2} & \sqrt{2}\sqrt{1+\sqrt{2}} \end{bmatrix},$$

resulting in the state feedback law

$$\mathbf{u}(t) = - \begin{bmatrix} 1+\sqrt{2} & \sqrt{2}\sqrt{1+\sqrt{2}} \end{bmatrix} \mathbf{x}(t).$$

The eigenvalues of the closed-loop system are (up to four digits) $\sigma(A - B R^{-1} B^T X_*) = \{-1.0987 \pm 0.4551i\}$.

3. The solution to the LQG problem for the inverted pendulum with Q, R as above and uncertainties $\mathbf{v}, \mathbf{w}$ with $\tilde{B} = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$ is obtained by combining the LQR solution derived above with the Kalman–Bucy filter obtained as in the examples part of the previous section. Thus, we get the LQG control law

$$\mathbf{u}(t) = - \begin{bmatrix} 1+\sqrt{2} & \sqrt{2}\sqrt{1+\sqrt{2}} \end{bmatrix} \hat{\mathbf{x}}(t),$$

where $\hat{\mathbf{x}}$ is the solution of

$$\dot{\hat{\mathbf{x}}}(t) = - \begin{bmatrix} 1+\sqrt{2} & -1 \\ 1+2\sqrt{2} & \sqrt{2}\sqrt{1+\sqrt{2}} \end{bmatrix} \hat{\mathbf{x}}(t) + (1+\sqrt{2}) \begin{bmatrix} 1 \\ 1 \end{bmatrix} \mathbf{y}(t).$$

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Fourier Analysis

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58.1 Introduction

Fourier analysis has been employed with great success in a wide range of applications. The underlying theory is based on a small set of linear transforms on particular linear spaces. It may, in fact, be best to refer to two parallel theories of Fourier analysis—the function/functional theory and the discrete theory—according to whether these linear spaces are infinite or finite dimensional. The function/functional theory involves infinite dimensional spaces of functions on $\mathbb{R}^n$ and can be further divided into two “subtheories”—one concerned with functions that are periodic on $\mathbb{R}^n$ (or can be treated as periodic extensions of functions on finite subregions) and one concerned with more general functions and functionals on $\mathbb{R}^n$. This theory played the major role in applications up to half a century or so ago. However, the tools from the discrete theory (involving vectors in $\mathbb{C}^N$ instead of functions on $\mathbb{R}^n$) are much more easily implemented on digital computers. This became of interest both because much of the function/functional theory analysis can be closely approximated within the discrete theory and because the discrete theory is a natural setting for functions known only by samplings. In addition, “fast” algorithms for computing the discrete transforms were developed, allowing discrete analysis to be done very quickly even with very large data sets. For these reasons the discrete theory has become extremely important in modern applications, and its utility, in turn, has greatly extended the applicability of Fourier analysis.

In the first part of this chapter, elements of the function/functional theory are presented and illustrated. For expediency, attention will be restricted to functions and functionals over subsets of $\mathbb{R}^1$. A corresponding review of the analogous elements of the discrete theory then follows, with a discussion of the relations between the two theories following that. Finally, one of the fast algorithms is described and the extent to which this algorithm improves speed and applicability is briefly discussed.

The development here is necessarily abbreviated and covers only a small fraction of the theory and applications of Fourier analysis. The reader interested in more complete treatments of the subject is encouraged to consult the references given throughout this chapter.